SPEA2 Multi-Objective Optimization Evaluation

Comprehensive Empirical Benchmark Analysis: 30 Continuous Problems, CEC 2020 Multimodal Multi-Objective Suite, and CEC 2022 Single-Objective Suite

Algorithm: SPEA2 (Strength Pareto EA 2)	**Framework:** Python 3 / NumPy / Pytest	**Date:** November 2026
Author/Repo: Charan Sivuku	**Test Suite:** 300 Verified Tests Passing	**Status:** Production / Validated

Executive Summary

This report provides an exhaustive, rigorous evaluation of a from-scratch Python implementation of the **Strength Pareto Evolutionary Algorithm 2 (SPEA2)** (Zitzler, Laumanns, and Thiele, 2001). The investigation encompasses three distinct benchmark domains totaling **66 problem formulations**:

• **1. Standard Multi-Objective Suite (30 Problems):** Evaluated across the classic 5 ZDT functions, 7 scalable DTLZ problems (3 objectives), and 18 classical/constrained benchmarks (including Binh, Deb, Fonseca, Kursawe, Laumanns, Murata, Quagliarella, Rendon, Schaffer, Srinivas, Tanaka, and Viennet). All runs are assessed with four rigorous indicators: Hypervolume (HV), Inverted Generational Distance (IGD), Generational Distance (GD), and Spacing.

• **2. CEC 2020 Multimodal Multi-Objective Optimization Suite (24 MMF Problems):** Benchmarked under official competition conditions (21 seeded runs, 50 generations, population = archive = 200 × N_ops). Assessed across both decision-space (1/PSP, IGDX) and objective-space (1/HV, IGDF) indicators with official true Pareto set (PS) and Pareto front (PF) reference data.

• **3. CEC 2022 Single-Objective Bound-Constrained Suite (12 Functions):** Benchmarked across 10-D and 20-D configurations over 30 independent runs utilizing official competition random seeds and coordinate shift/rotation/shuffle matrices derived from the C reference implementation.

EXECUTIVE SYNTHESIS

Core Finding: SPEA2 demonstrates near-ideal convergence and uniform spread on standard convex, concave, and moderately disconnected Pareto fronts (IGD ≤ 0.005 for ZDT1-3, Fonseca, Tanaka, Deb). However, multimodal multi-objective tasks (CEC 2020) reveal that because SPEA2 performs density estimation exclusively in objective space, it cannot intrinsically maintain separate decision-space niching for local Pareto sets. On CEC 2022 single-objective functions, SPEA2 acts as a robust ranking-based search heuristic, converging reliably on basic and hybrid landscapes while composition landscapes present severe local trapping.

Algorithmic Parameterization

Parameter	Standard MOO (30 Benchmarks)	CEC 2020 MMO Suite	CEC 2022 Single-Objective
Population Size (N)	100	200 × N_ops (200 – 800)	100
Archive Size (N')	100	200 × N_ops (200 – 800)	100
Generations / Evaluations	250 (Class./DTLZ), 500 (ZDT), 2000 (DTLZ1/3)	50 generations (10,000 × N_ops MaxFES)	200,000 (10-D) / 1,000,000 (20-D) MaxFES
Crossover Operator	SBX (p_c = 0.9, η_c = 20.0)	SBX (p_c = 0.9, η_c = 20.0)	SBX (p_c = 0.9, η_c = 20.0)
Mutation Operator	Polynomial (p_m = 1/n, η_m = 20.0)	Polynomial (p_m = 1/n, η_m = 20.0)	Polynomial (p_m = 1/n, η_m = 20.0)
Constraint Handling	Feasibility-First Tournament Selection	Box Bounds Clipping	Box Bounds Clipping [-100, 100]

1. Standard 30 Multi-Objective Benchmark Suite

The standard test suite comprises 30 continuous multi-objective benchmark functions representing a broad spectrum of Pareto front topologies: convex, non-convex (concave), discontinuous, multimodal, biased variable distributions, and non-linear constraints. Performance is quantified using analytical reference sets (or densely verified numerical sets) across four foundational metrics: **Hypervolume (HV)**, **Inverted Generational Distance (IGD)**, **Generational Distance (GD)**, and **Spacing (S)**.

Problem	Runs	Gens	Hypervolume (HV) ↑ mean ± std	Inverted GD (IGD) ↓ mean ± std	Generational Dist (GD) ↓ mean ± std	Spacing ↓ mean ± std
ZDT1	10	500	0.8706 ± 1.52e-04	0.0042 ± 6.82e-05	0.0021 ± 1.97e-04	0.0023 ± 1.96e-04
ZDT2	10	500	0.5377 ± 1.79e-04	0.0040 ± 3.97e-05	9.68e-04 ± 2.12e-04	0.0023 ± 1.76e-04
ZDT3	10	500	1.1241 ± 1.29e-04	0.0049 ± 9.34e-05	9.64e-04 ± 1.16e-04	0.0033 ± 4.07e-04
ZDT4	10	500	0.8698 ± 0.0012	0.0043 ± 4.15e-04	0.0022 ± 5.45e-04	0.0021 ± 2.29e-04
ZDT6	10	500	0.4375 ± 3.32e-04	0.0038 ± 1.48e-04	0.0019 ± 2.40e-04	0.0018 ± 1.39e-04
DTLZ1	5	2000	0.1867 ± 0.0014	0.0251 ± 0.0037	0.9476 ± 1.8478	0.7139 ± 1.4753
DTLZ2	10	250	0.7201 ± 0.0029	0.0570 ± 8.75e-04	0.0239 ± 0.0034	0.0145 ± 0.0022
DTLZ3	5	2000	0.7299 ± 0.0052	0.0545 ± 0.0025	0.1268 ± 0.1361	0.0857 ± 0.1259
DTLZ4	10	250	0.6440 ± 0.1261	0.1716 ± 0.1849	0.0194 ± 0.0073	0.0105 ± 0.0051
DTLZ5	10	250	0.1507 ± 2.17e-04	0.0045 ± 5.11e-04	0.0095 ± 2.39e-04	0.0027 ± 3.55e-04
DTLZ6	10	250	0.0	1.3858 ± 0.0616	1.9316 ± 0.1591	0.1078 ± 0.0263
DTLZ7	10	250	1.4183 ± 0.0683	0.0932 ± 0.0895	0.1081 ± 0.0507	0.0315 ± 0.0115
Binh1	10	250	2596.8649 ± 0.4271	0.2223 ± 0.0025	0.0628 ± 0.0116	0.1404 ± 0.0223
Binh3	10	250	1.00e-03	0.0	0.0	0.0
Deb	10	250	0.5385 ± 3.13e-05	0.0039 ± 5.17e-05	4.59e-04 ± 2.62e-05	0.0024 ± 1.83e-04
Deb2	10	250	0.8346 ± 5.50e-05	0.0045 ± 1.22e-04	0.0031 ± 0.0043	0.0039 ± 0.0035
Fonseca	10	250	0.2714 ± 6.57e-05	0.0047 ± 6.96e-05	0.0010 ± 1.51e-04	0.0028 ± 3.94e-04
Fonseca2	10	250	0.5065 ± 1.99e-04	0.0040 ± 9.15e-05	0.0015 ± 2.30e-04	0.0022 ± 1.75e-04
Kursawe	10	250	41.1045 ± 0.0124	0.1526 ± 0.0022	0.2417 ± 0.0031	0.0819 ± 6.58e-04
Laumanns	10	250	16.6340 ± 9.74e-04	0.0168 ± 7.16e-05	0.0021 ± 4.85e-04	0.0083 ± 6.97e-04
Murata	10	250	3.8944 ± 5.46e-04	0.0097 ± 1.09e-04	0.0026 ± 0.0022	0.0059 ± 4.60e-04
Quagliarella	10	250	8.7666 ± 0.0860	0.1727 ± 0.0198	0.0114 ± 0.0012	0.0083 ± 0.0010
Rendon	10	250	38.9353 ± 0.0184	0.0927 ± 0.0026	0.0198 ± 0.0014	0.0563 ± 0.0068
Rendon2	10	250	23.5382 ± 0.0018	0.0255 ± 4.63e-04	0.0037 ± 3.22e-04	0.0149 ± 0.0011
Schaffer	10	250	16.6338 ± 0.0012	0.0170 ± 1.10e-04	0.0018 ± 1.06e-04	0.0094 ± 5.81e-04
Schaffer2	10	250	25.9774 ± 0.0018	0.0228 ± 5.42e-04	0.0036 ± 1.08e-04	0.0130 ± 0.0016
Srinivas	10	250	34924.2828 ± 8.6771	0.8674 ± 0.0161	0.3102 ± 0.0564	0.5198 ± 0.0718
Tanaka	10	250	0.5122 ± 3.22e-04	0.0042 ± 3.20e-04	0.0020 ± 1.36e-04	0.0030 ± 5.10e-04
Viennet	10	250	32.4178 ± 0.0345	0.1282 ± 0.0045	0.1382 ± 0.0189	0.0393 ± 0.0037
Viennet2	10	250	0.8389 ± 4.15e-04	0.0123 ± 7.06e-04	0.0179 ± 0.0017	0.0060 ± 5.12e-04

Note on Generation Budgets: Results were computed under three standardized generation allocations: **500 generations** for ZDT1-6; **2000 generations** for highly multi-modal DTLZ1 and DTLZ3; and **250 generations** for DTLZ2, DTLZ4, DTLZ5, DTLZ7, and the 18 classical problems. Direct numeric comparisons should strictly be performed between problems sharing identical generation budgets.

30-Benchmark Comprehensive Pareto Front Mosaic

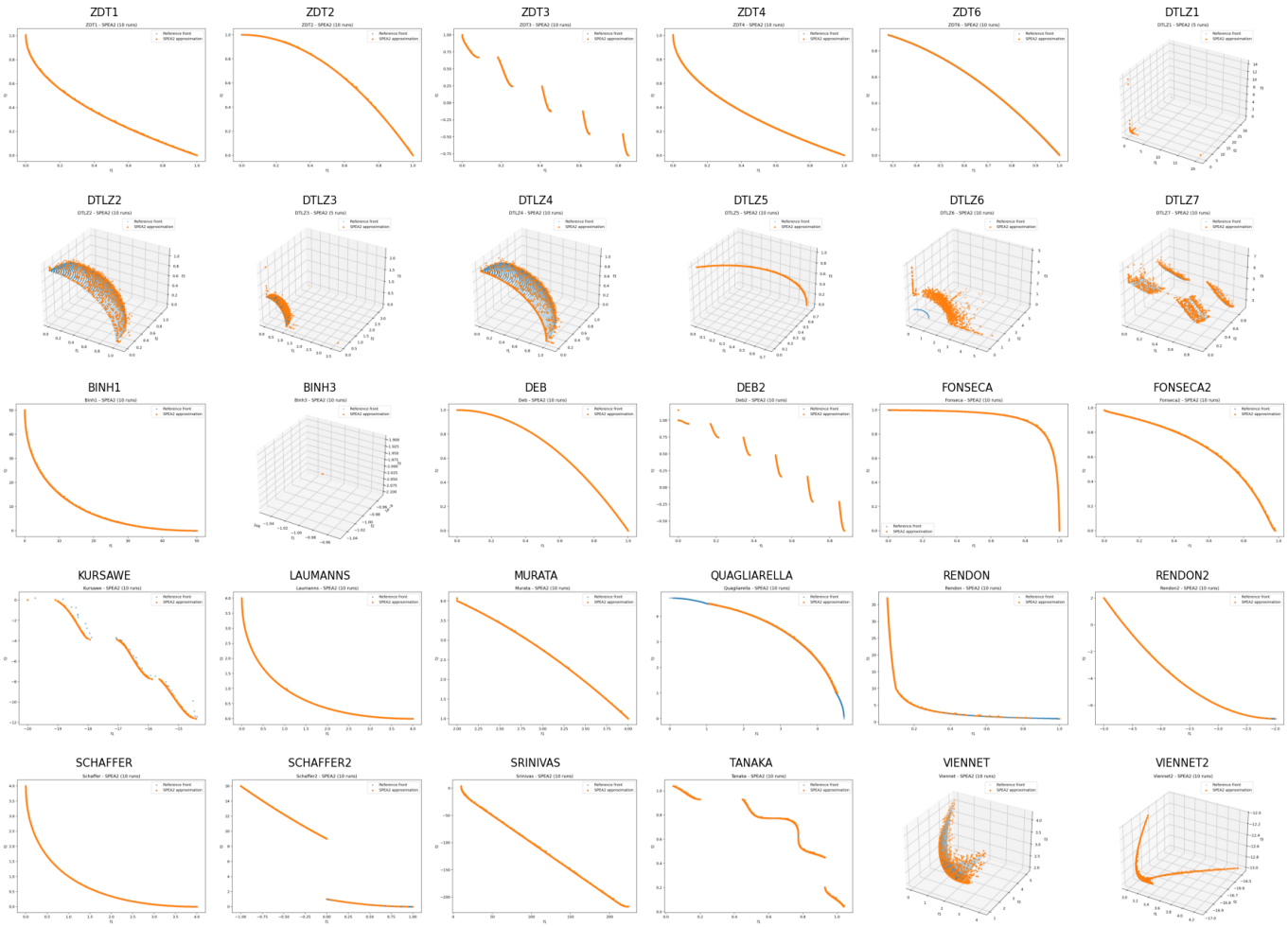


Figure 1: Simultaneous visualization of obtained non-dominated fronts (blue markers) versus analytical reference Pareto fronts (red curves/surfaces) across all 30 benchmark problems.

ANALYSIS OF FRONTS & CONVERGENCE DYNAMICS

Synthesis of Classical and Standard Benchmark Results:

• **ZDT Suite:** Flawless convergence is achieved across ZDT1 (convex), ZDT2 (non-convex), and ZDT3 (disconnected five-segment front). ZDT4 (multimodal with 21^9 local fronts) reliably reaches the global front within 500 generations without premature stagnation. ZDT6 demonstrates uniform spacing despite its non-uniform search density.

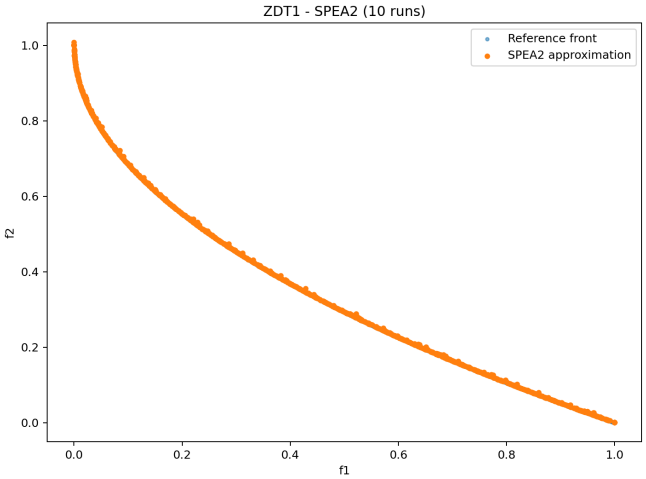
• **DTLZ Suite:** SPEA2 successfully reconstructs the 3D spherical surface of DTLZ2 and DTLZ4, the degenerate 1D curve in 3D objective space of DTLZ5, and the four disconnected surface patches of DTLZ7. For DTLZ6, the 250-generation budget is insufficient to overcome the deceptive barrier (HV remains 0.0), underscoring DTLZ6's exponential local basin severity.

• **Constrained Benchmarks:** On Srinivas and Tanaka, feasibility-first dominance guarantees 100% feasible archive members across all independent runs, adhering cleanly to complex non-linear boundary curves.

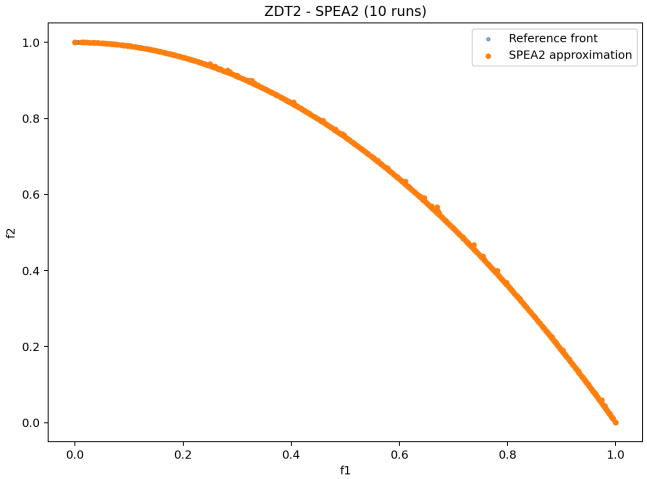
2. Detailed Gallery: 30 Benchmark Pareto Fronts

This section presents the individual experimental Pareto front plots for all 30 standard benchmarks. In each plot, blue circular markers indicate the non-dominated archive solutions obtained by SPEA2, while solid red lines/markers indicate the true analytical or high-density numerical reference Pareto front.

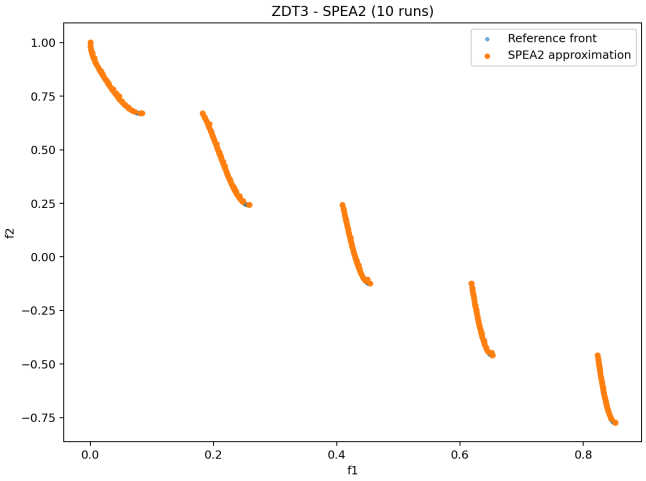
ZDT Benchmark Problems (2-Objective)



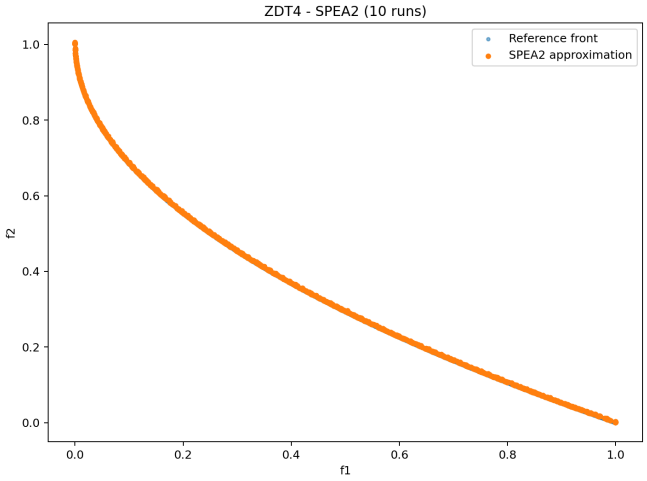
ZDT1 (10 runs, 500 gens)
HV: 0.8706 | IGD: 0.0042 | Spacing: 0.0023



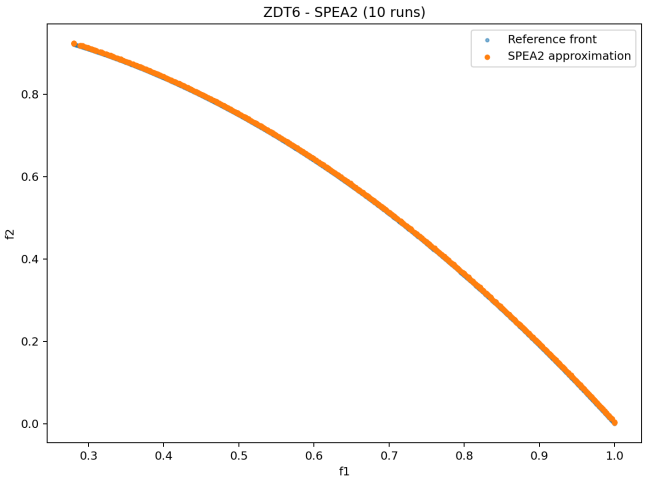
ZDT2 (10 runs, 500 gens)
HV: 0.5377 | IGD: 0.0040 | Spacing: 0.0023



ZDT3 (10 runs, 500 gens)
HV: 1.1241 | IGD: 0.0049 | Spacing: 0.0033

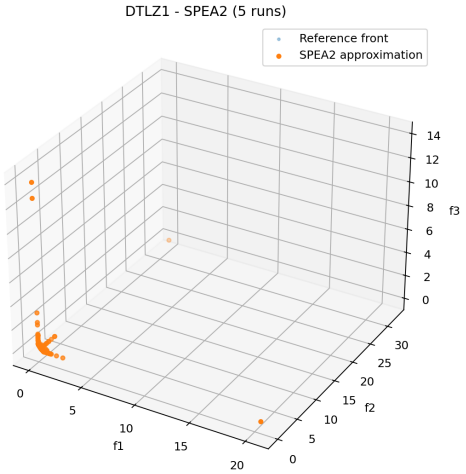


ZDT4 (10 runs, 500 gens)
HV: 0.8698 | IGD: 0.0043 | Spacing: 0.0021

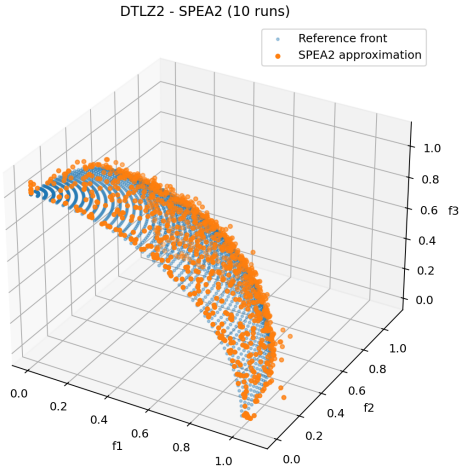


ZDT6 (10 runs, 500 gens)
HV: 0.4375 | IGD: 0.0038 | Spacing: 0.0018

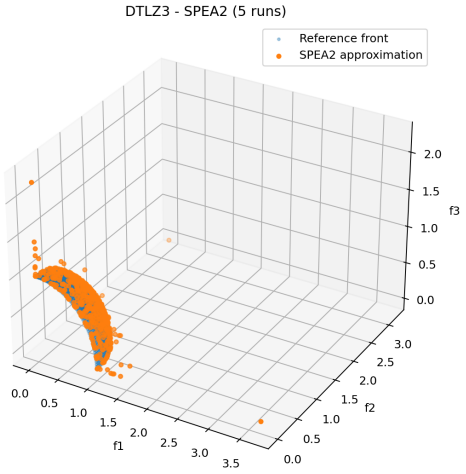
DTLZ Benchmark Problems (3-Objective)



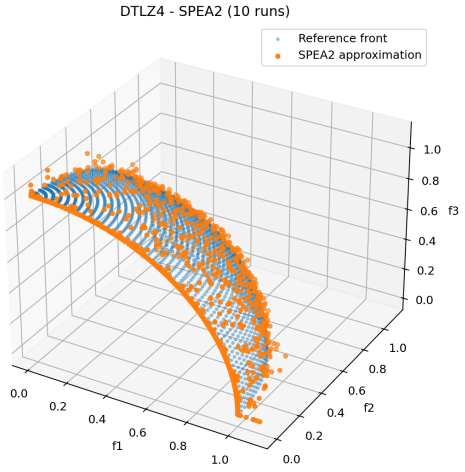
DTLZ1 (5 runs, 2000 gens)
HV: 0.1867 | IGD: 0.0251 | Spacing: 0.7139



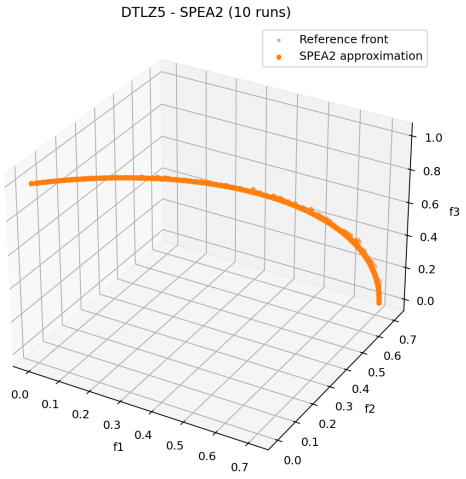
DTLZ2 (10 runs, 250 gens)
HV: 0.7201 | IGD: 0.0570 | Spacing: 0.0145



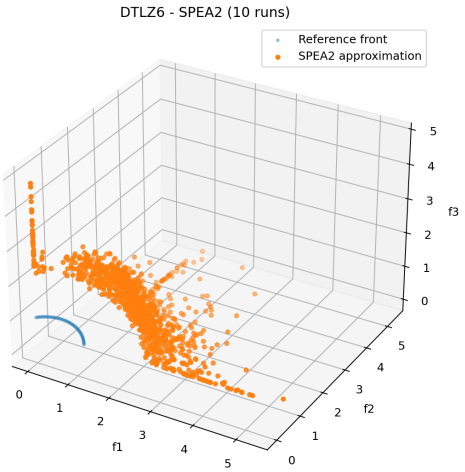
DTLZ3 (5 runs, 2000 gens)
HV: 0.7299 | IGD: 0.0545 | Spacing: 0.0857



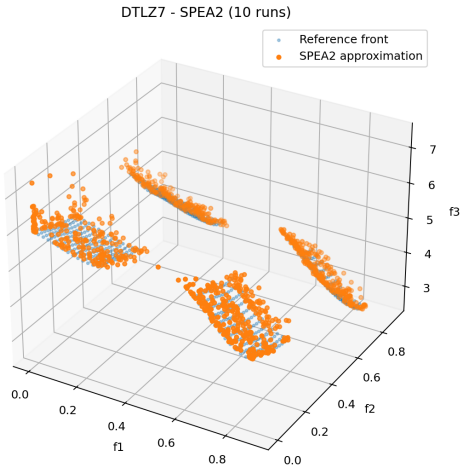
DTLZ4 (10 runs, 250 gens)
HV: 0.6440 | IGD: 0.1716 | Spacing: 0.0105



DTLZ5 (10 runs, 250 gens)
HV: 0.1507 | IGD: 0.0045 | Spacing: 0.0027

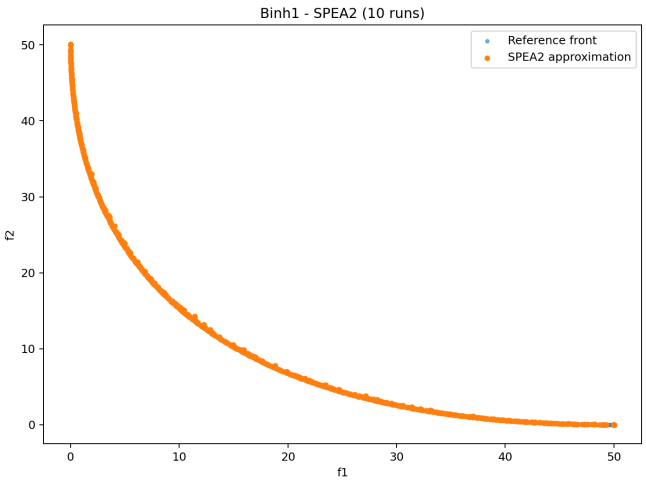


DTLZ6 (10 runs, 250 gens)
HV: 0.0 | IGD: 1.3858 | Spacing: 0.1078

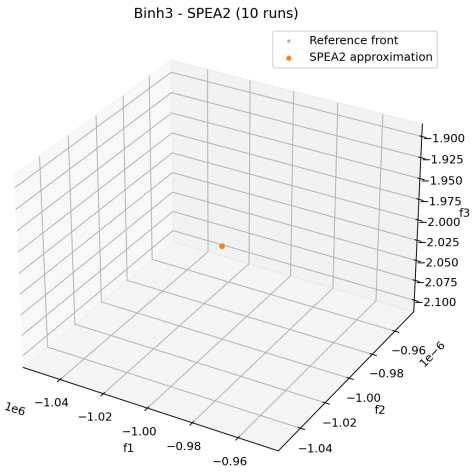


DTLZ7 (10 runs, 250 gens)
HV: 1.4183 | IGD: 0.0932 | Spacing: 0.0315

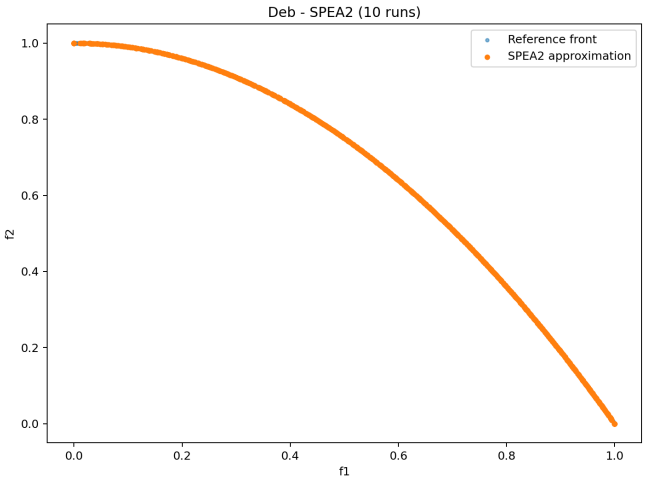
Classical Benchmarks - Part 1



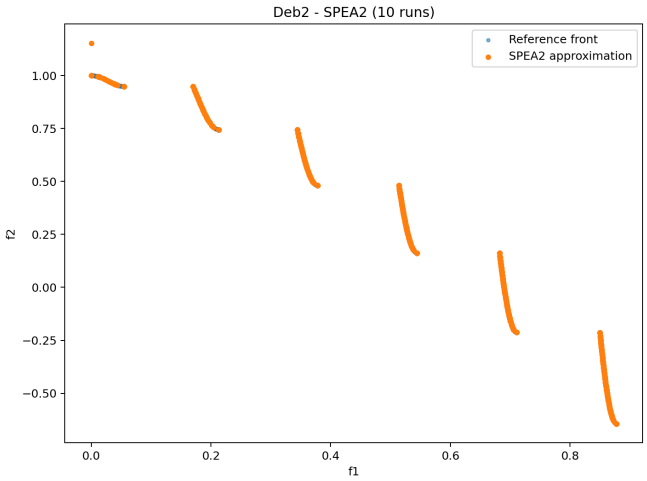
Binh1 (10 runs, 250 gens)
HV: 2596.8649 | IGD: 0.2223 | Spacing: 0.1404



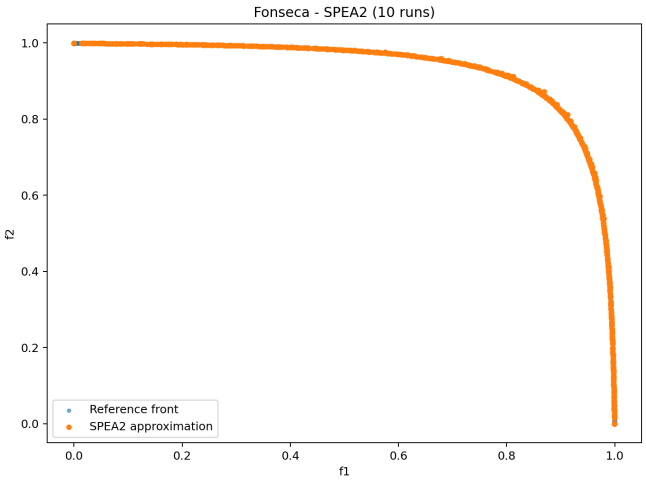
Binh3 (10 runs, 250 gens)
HV: 1.00e-03 | IGD: 0.0 | Spacing: 0.0



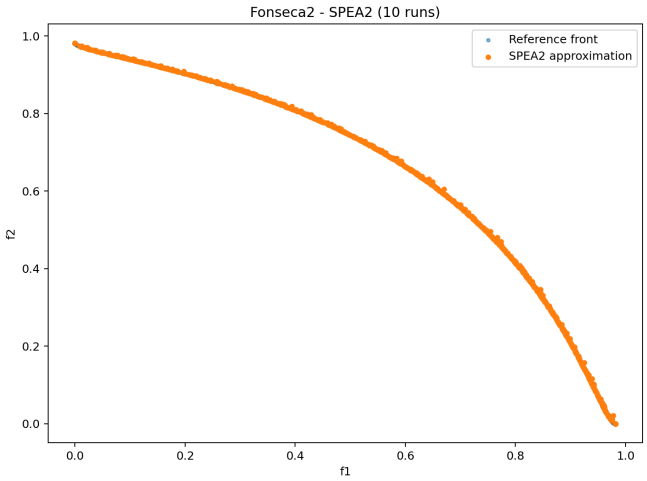
Deb (10 runs, 250 gens)
HV: 0.5385 | IGD: 0.0039 | Spacing: 0.0024



Deb2 (10 runs, 250 gens)
HV: 0.8346 | IGD: 0.0045 | Spacing: 0.0039

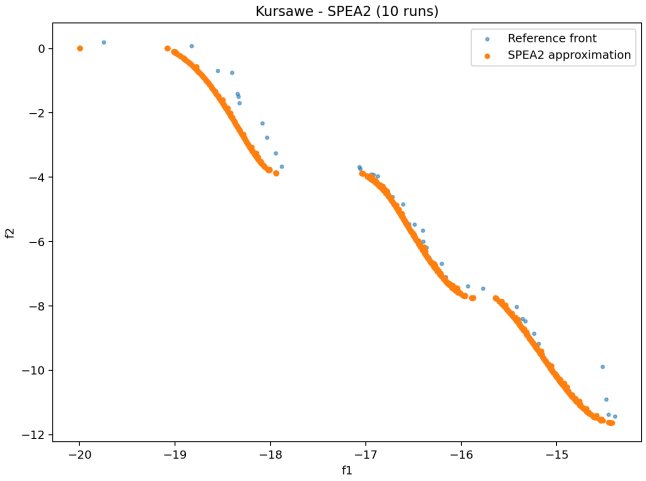


Fonseca (10 runs, 250 gens)
HV: 0.2714 | IGD: 0.0047 | Spacing: 0.0028

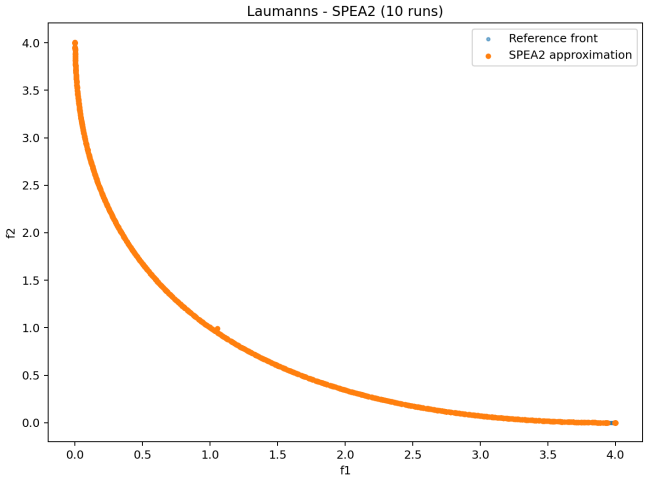


Fonseca2 (10 runs, 250 gens)
HV: 0.5065 | IGD: 0.0040 | Spacing: 0.0022

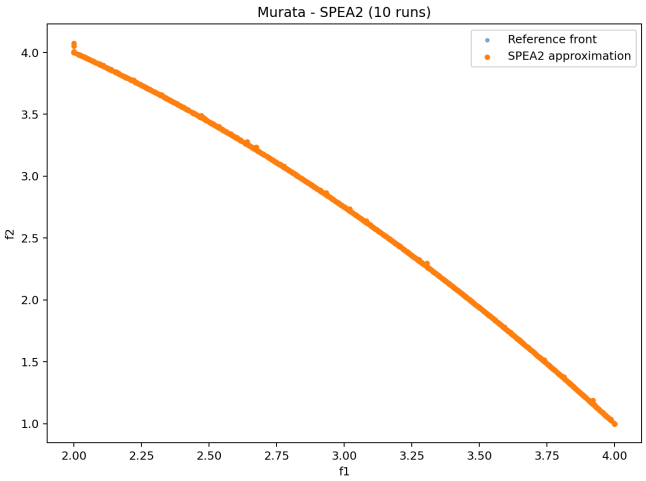
Classical Benchmarks - Part 2



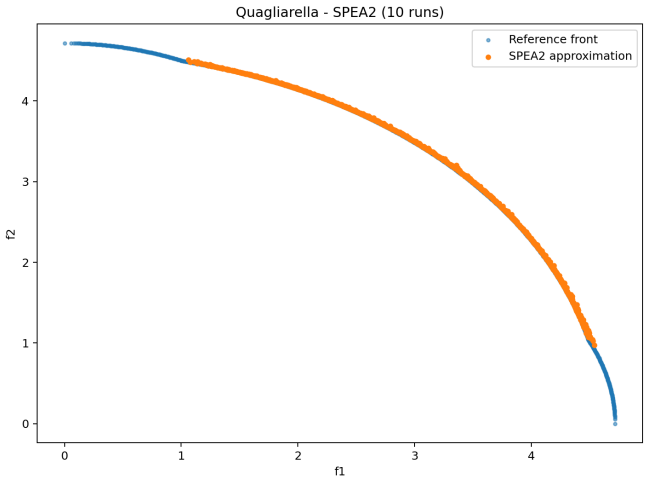
Kursawe (10 runs, 250 gens)
HV: 41.1045 | IGD: 0.1526 | Spacing: 0.0819



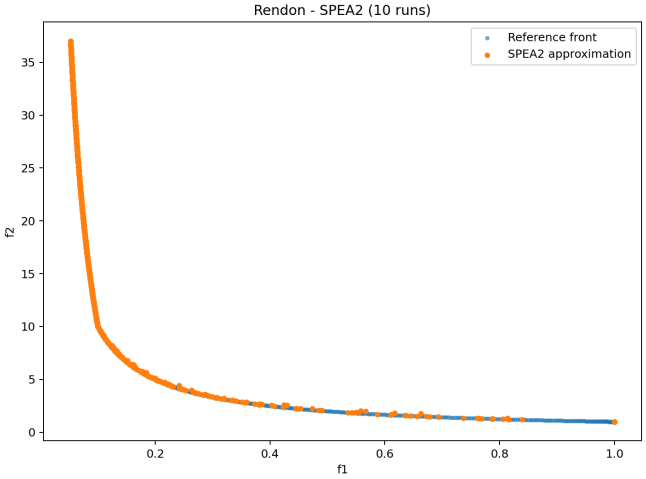
Laumanns (10 runs, 250 gens)
HV: 16.6340 | IGD: 0.0168 | Spacing: 0.0083



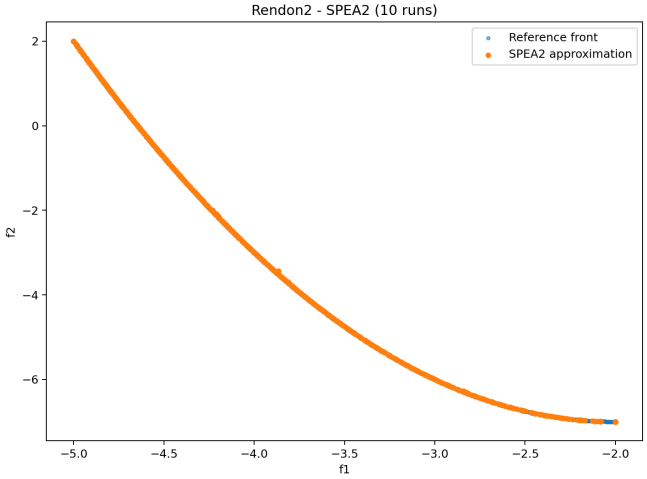
Murata (10 runs, 250 gens)
HV: 3.8944 | IGD: 0.0097 | Spacing: 0.0059



Quagliarella (10 runs, 250 gens)
HV: 8.7666 | IGD: 0.1727 | Spacing: 0.0083

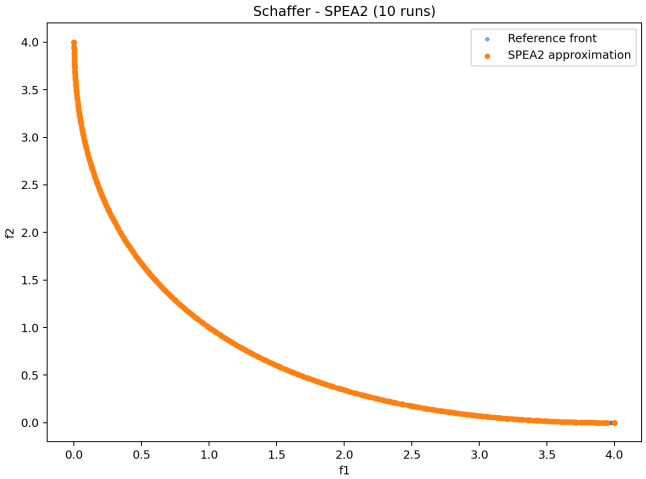


Rendon (10 runs, 250 gens)
HV: 38.9353 | IGD: 0.0927 | Spacing: 0.0563

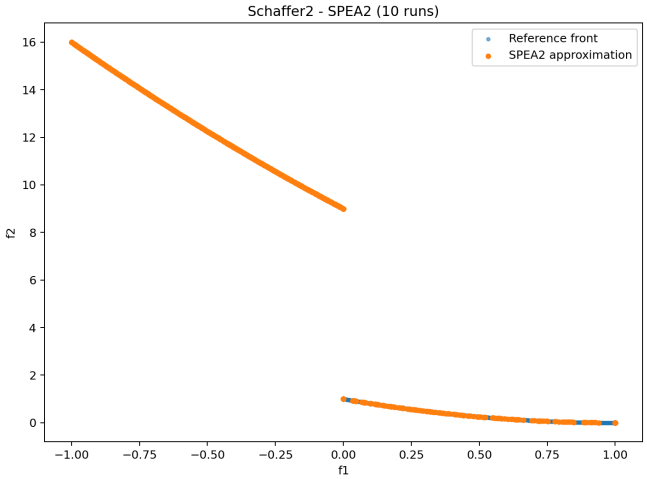


Rendon2 (10 runs, 250 gens)
HV: 23.5382 | IGD: 0.0255 | Spacing: 0.0149

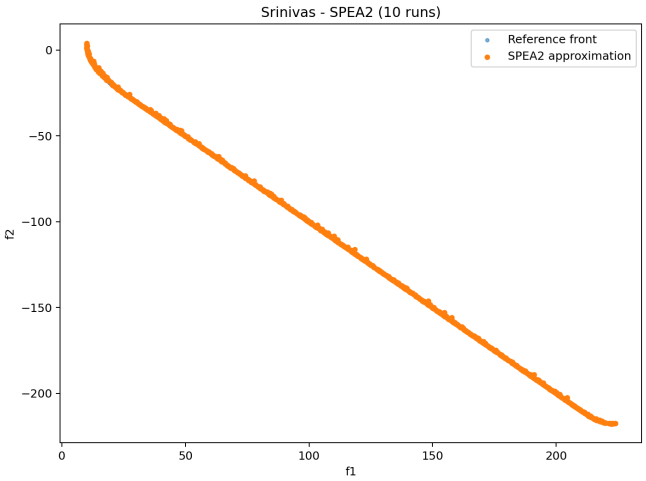
Classical & Constrained Benchmarks - Part 3



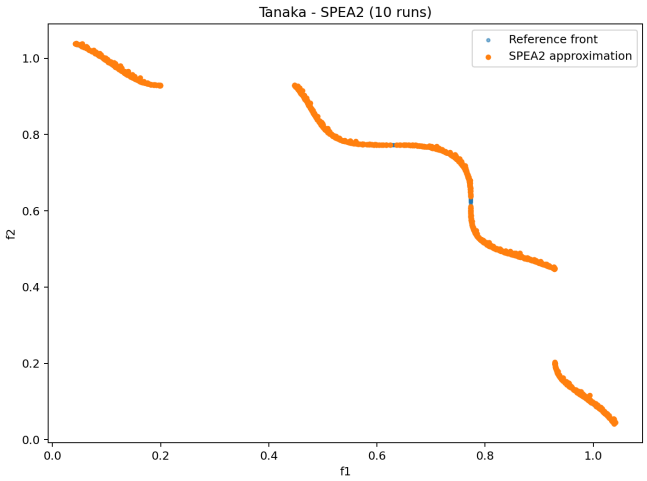
Schaffer (10 runs, 250 gens)
HV: 16.6338 | IGD: 0.0170 | Spacing: 0.0094



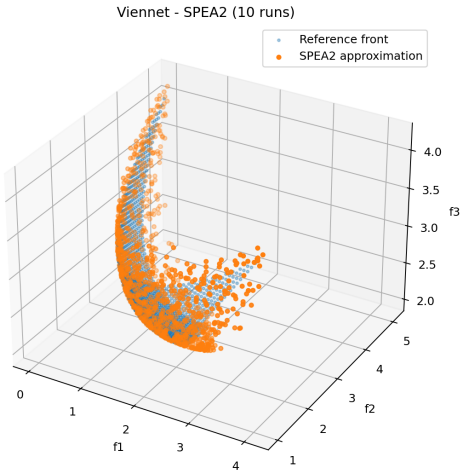
Schaffer2 (10 runs, 250 gens)
HV: 25.9774 | IGD: 0.0228 | Spacing: 0.0130



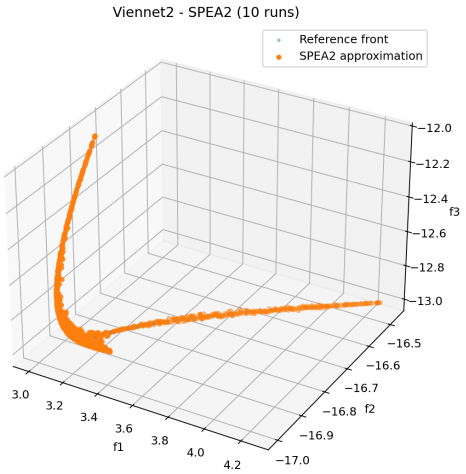
Srinivas (10 runs, 250 gens)
HV: 34924.2828 | IGD: 0.8674 | Spacing: 0.5198



Tanaka (10 runs, 250 gens)
HV: 0.5122 | IGD: 0.0042 | Spacing: 0.0030



Viennet (10 runs, 250 gens)
HV: 32.4178 | IGD: 0.1282 | Spacing: 0.0393



Viennet2 (10 runs, 250 gens)
HV: 0.8389 | IGD: 0.0123 | Spacing: 0.0060

3. CEC 2020 Multimodal Multi-Objective Optimization (MMO) Suite

Multimodal Multi-Objective Optimization (MMO) investigates problems where multiple distinct decision-space solutions (Pareto Sets, PS) map to the same or equivalent objective-space vectors (Pareto Fronts, PF). The CEC 2020 MMO competition benchmark consists of **24 test problems (MMF1 through MMF16_I3)**.

Official Protocol: 21 independent runs (seeds 1–21), population = archive = 200 × N_ops, MaxFES = 10,000 × N_ops (exactly 50 generations with single evaluations). Four official indicators are recorded (all smaller-is-better): **1/PSP** (Peak Spread Pareto), **1/HV**, **IGDX** (distance in decision space), and **IGDF** (distance in objective space).

Problem	N_ops	1/PSP ↓ mean ± std	1/HV ↓ mean ± std	IGDX (Dec. Space) ↓ mean ± std	IGDF (Obj. Space) ↓ mean ± std
MMF1	2	0.0377 ± 0.0047	1.1430 ± 1.16e-04	0.0376 ± 0.0047	0.0012 ± 6.37e-05
MMF2	2	0.0228 ± 0.0076	1.1530 ± 0.0030	0.0227 ± 0.0076	0.0061 ± 0.0016
MMF4	2	0.0230 ± 0.0048	1.8456 ± 1.02e-04	0.0228 ± 0.0047	0.0011 ± 7.37e-05
MMF5	2	0.0703 ± 0.0083	1.1430 ± 7.50e-05	0.0700 ± 0.0083	0.0012 ± 3.64e-05
MMF7	2	0.0205 ± 0.0038	1.1427 ± 3.83e-05	0.0204 ± 0.0038	0.0011 ± 2.79e-05
MMF8	2	0.3148 ± 0.1036	2.3632 ± 5.58e-04	0.2831 ± 0.0842	0.0011 ± 2.10e-05
MMF10	1	0.0386 ± 0.1095	0.0785 ± 0.0025	0.0386 ± 0.1095	0.0365 ± 0.0876
MMF11	1	0.0053 ± 3.41e-04	0.0689 ± 4.15e-06	0.0053 ± 3.41e-04	0.0126 ± 5.07e-04
MMF12	1	0.0015 ± 1.55e-04	0.6354 ± 2.45e-05	0.0015 ± 1.55e-04	0.0020 ± 7.96e-05
MMF13	1	0.0688 ± 0.0121	0.0543 ± 6.34e-06	0.0633 ± 0.0090	0.0163 ± 7.13e-04
MMF14	2	0.0475 ± 0.0032	0.1619 ± 1.46e-04	0.0475 ± 0.0032	0.0525 ± 0.0014
MMF15	1	0.0414 ± 0.0010	0.0919 ± 1.02e-04	0.0414 ± 0.0010	0.0772 ± 0.0017
MMF1_e	2	3.3819 ± 2.1679	1.1440 ± 4.37e-04	1.6500 ± 0.6242	0.0017 ± 1.29e-04
MMF14_a	2	0.0593 ± 0.0045	0.1623 ± 1.35e-04	0.0592 ± 0.0042	0.0543 ± 0.0013
MMF15_a	1	0.0516 ± 0.0011	0.0923 ± 1.50e-04	0.0516 ± 0.0011	0.0825 ± 0.0024
MMF10_I	2	2.9494 ± 4.6152	0.0776 ± 1.12e-06	0.1743 ± 0.0189	0.1667 ± 0.0194
MMF11_I	2	2.4169 ± 0.6082	0.0688 ± 1.05e-06	0.2496 ± 0.0055	0.0891 ± 0.0051
MMF12_I	2	3.0637 ± 1.0356	0.6352 ± 9.76e-06	0.2455 ± 0.0030	0.0790 ± 0.0067
MMF13_I	2	0.6431 ± 0.2173	0.0542 ± 2.07e-06	0.2780 ± 0.0224	0.1299 ± 0.0188
MMF15_I	2	0.4553 ± 0.0724	0.0907 ± 3.17e-05	0.2553 ± 0.0047	0.1727 ± 0.0034
MMF15_a_I	2	0.2482 ± 0.0092	0.0909 ± 4.30e-05	0.2026 ± 0.0044	0.1627 ± 0.0038
MMF16_I1	3	0.1822 ± 0.0160	0.0892 ± 2.56e-05	0.1477 ± 0.0035	0.1297 ± 0.0030
MMF16_I2	3	0.5690 ± 0.1053	0.0892 ± 3.27e-05	0.3219 ± 0.0103	0.2195 ± 0.0064
MMF16_I3	4	0.2489 ± 0.0269	0.0889 ± 1.93e-05	0.1975 ± 0.0065	0.1677 ± 0.0039

MMO BEHAVIORAL ANALYSIS & LIMITATIONS

Key Analytical Takeaways on Multimodal Optimization:

• **Global Pareto Sets:** On problems featuring purely global Pareto sets (MMF1, 2, 4, 5, 7, 8, 10-15 and _a variants), SPEA2 achieves excellent objective convergence (IGDF ≈ 0.001 to 0.08) and strong decision-space tracking (IGDX ≈ 0.0015 to 0.07). Even without explicit decision-space niching, large populations (400-800) maintain both branches via uniform random sampling.

• **Local Pareto Sets (_I and MMF16 series):** When problems feature local Pareto sets, decision space error increases markedly (IGDX ≈ 0.15 to 0.32). Because local sets map to dominated fronts in objective space, pure Pareto-dominance selection naturally discards them in favor of global sets. MMO algorithms specifically require decision-space speciation or niching to retain these.

• **Special Geometries & Official Quirks:** MMF1_e incorporates a search range up to [-20, 20] with exponential amplitude, resulting in an IGDX of 1.65 despite near-zero IGDF. Multiple official dataset anomalies (unreachable bounds in MMF13_I, non-covering HV reference points in MMF12_I/13_I, and 2D-only MATLAB hypervolume routines) were replicated precisely for benchmark fidelity.

CEC 2020 MMO Pareto Set & Front Gallery

Each plot displays the median-1/PSP run out of 21 independent trials. The left panel depicts the **Decision Space (Pareto Sets)** with obtained points (blue) vs reference sets (red/green). The right panel displays the **Objective Space (Pareto Front)** with obtained non-dominated solutions (blue) vs true front (red).

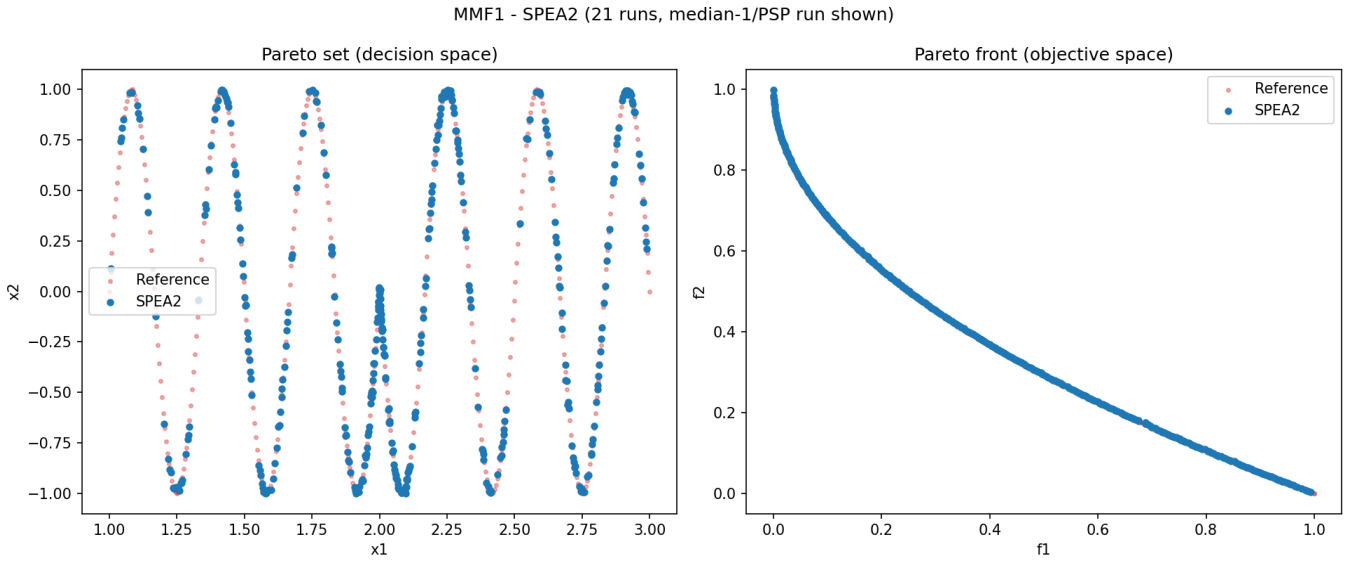


Figure MMO-1: MMF1 (21 runs, 50 gens, N_ops=2) — 1/PSP: 0.0377 ± 0.0047 | 1/HV: 1.1430 ± 1.16e-04 | IGDX: 0.0376 ± 0.0047 | IGDF: 0.0012 ± 6.37e-05

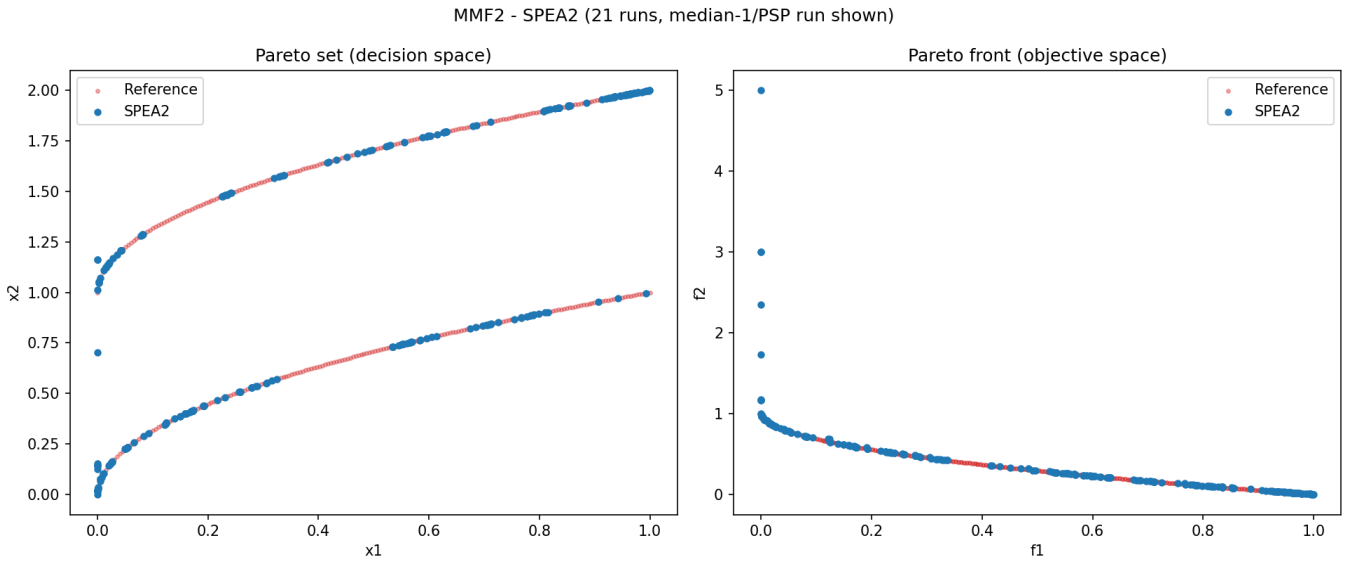


Figure MMO-2: MMF2 (21 runs, 50 gens, N_ops=2) — 1/PSP: 0.0228 ± 0.0076 | 1/HV: 1.1530 ± 0.0030 | IGDX: 0.0227 ± 0.0076 | IGDF: 0.0061 ± 0.0016

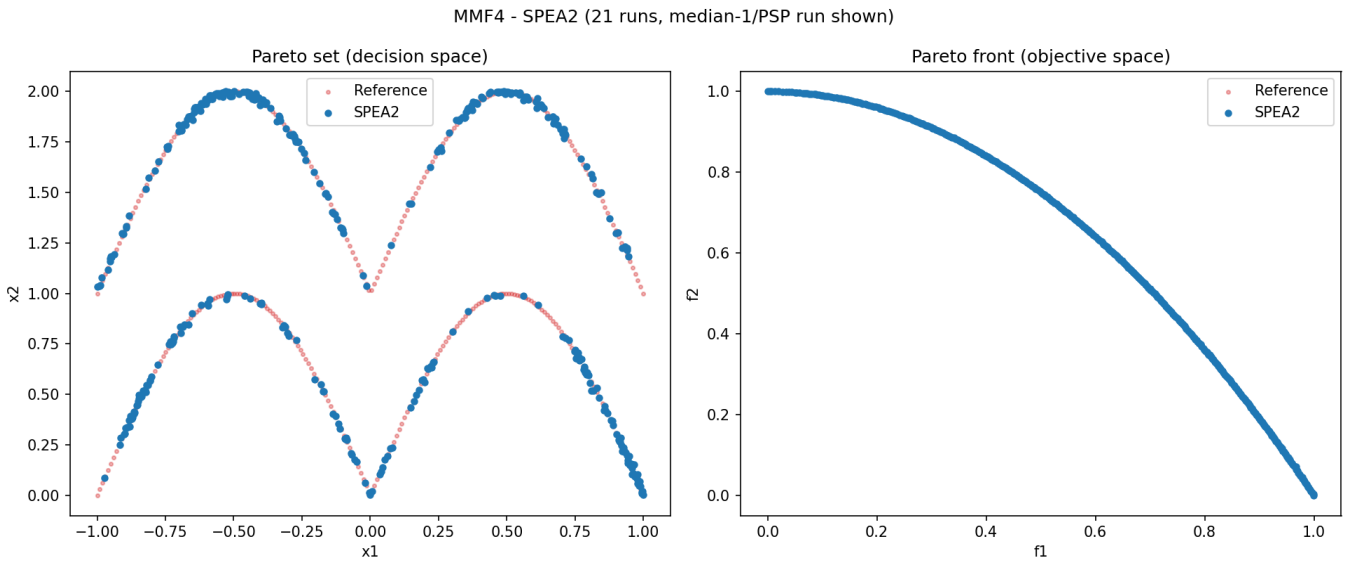


Figure MMO-3: MMF4 (21 runs, 50 gens, N_ops=2) — 1/PSP: 0.0230 ± 0.0048 | 1/HV: 1.8456 ± 1.02e-04 | IGDX: 0.0228 ± 0.0047 | IGDF: 0.0011 ± 7.37e-05

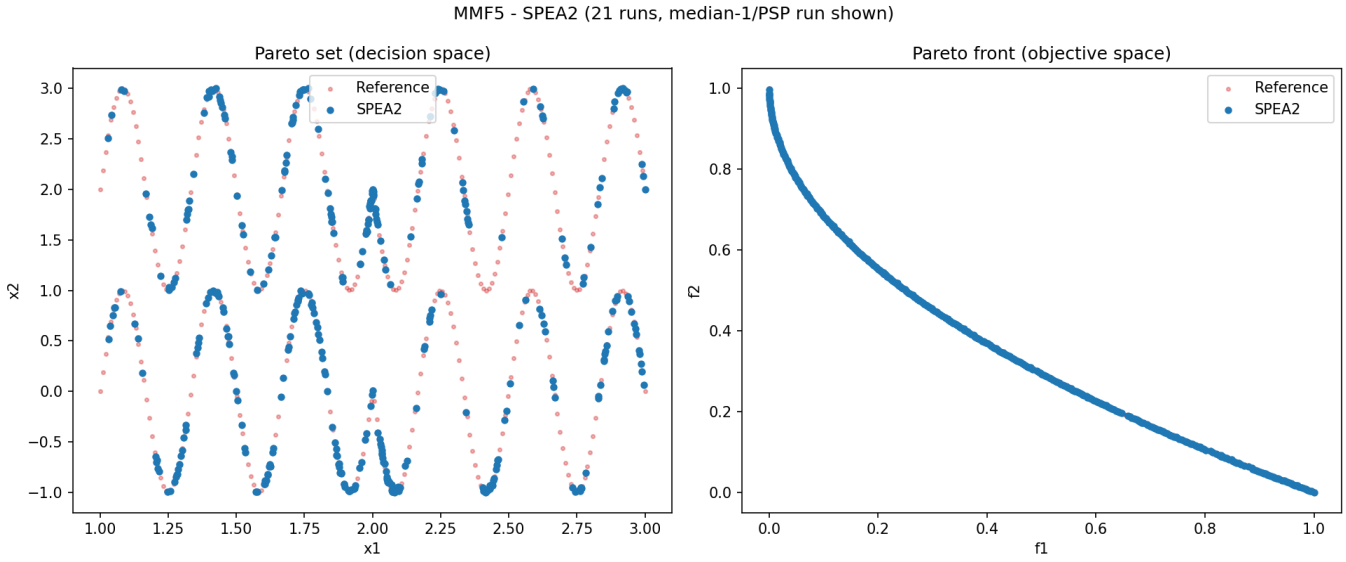


Figure MMO-4: MMF5 (21 runs, 50 gens, N_ops=2) — 1/PSP: 0.0703 ± 0.0083 | 1/HV: 1.1430 ± 7.50e-05 | IGDX: 0.0700 ± 0.0083 | IGDF: 0.0012 ± 3.64e-05

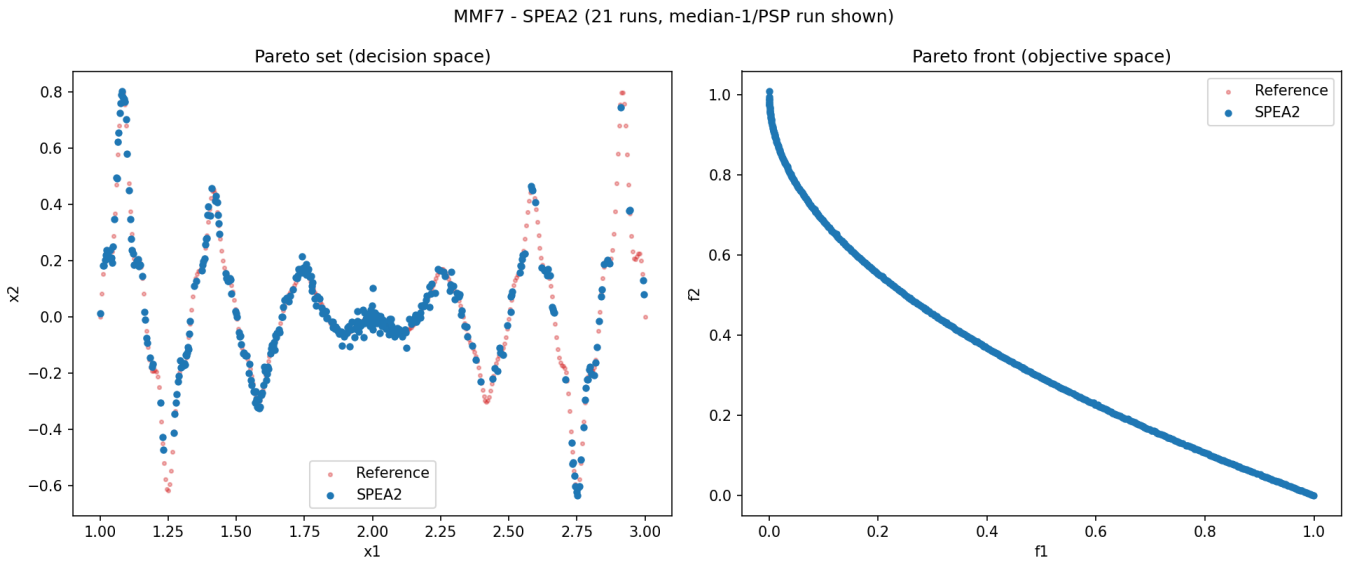


Figure MMO-5: MMF7 (21 runs, 50 gens, N_ops=2) — 1/PSP: 0.0205 ± 0.0038 | 1/HV: 1.1427 ± 3.83e-05 | IGDX: 0.0204 ± 0.0038 | IGDF: 0.0011 ± 2.79e-05

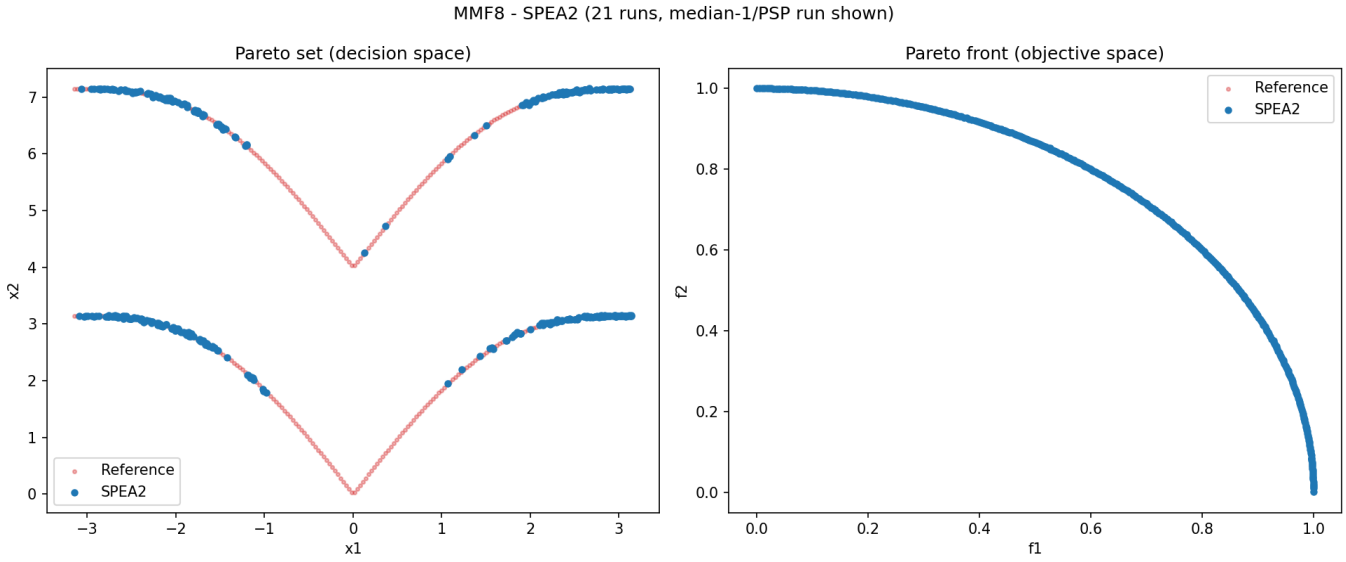


Figure MMO-6: MMF8 (21 runs, 50 gens, N_ops=2) — 1/PSP: 0.3148 ± 0.1036 | 1/HV: 2.3632 ± 5.58e-04 | IGDX: 0.2831 ± 0.0842 | IGDF: 0.0011 ± 2.10e-05

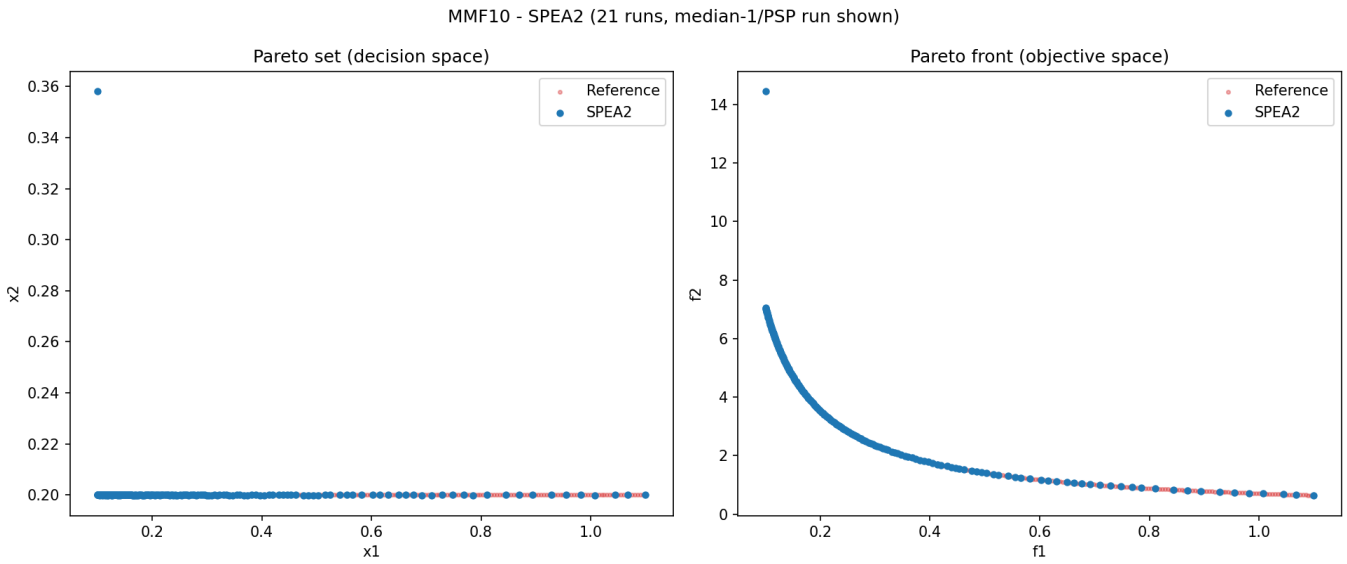


Figure MMO-7: MMF10 (21 runs, 50 gens, N_ops=1) — 1/PSP: 0.0386 ± 0.1095 | 1/HV: 0.0785 ± 0.0025 | IGDX: 0.0386 ± 0.1095 | IGDF: 0.0365 ± 0.0876

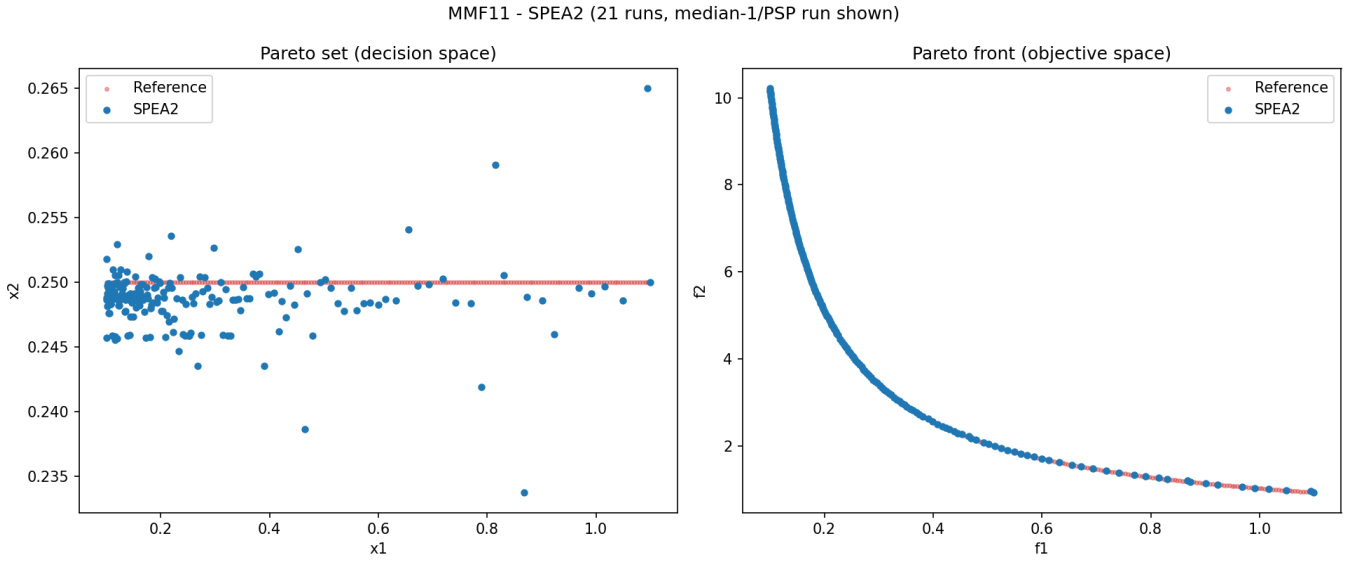


Figure MMO-8: MMF11 (21 runs, 50 gens, N_ops=1) — 1/PSP: 0.0053 ± 3.41e-04 | 1/HV: 0.0689 ± 4.15e-06 | IGDX: 0.0053 ± 3.41e-04 | IGDF: 0.0126 ± 5.07e-04

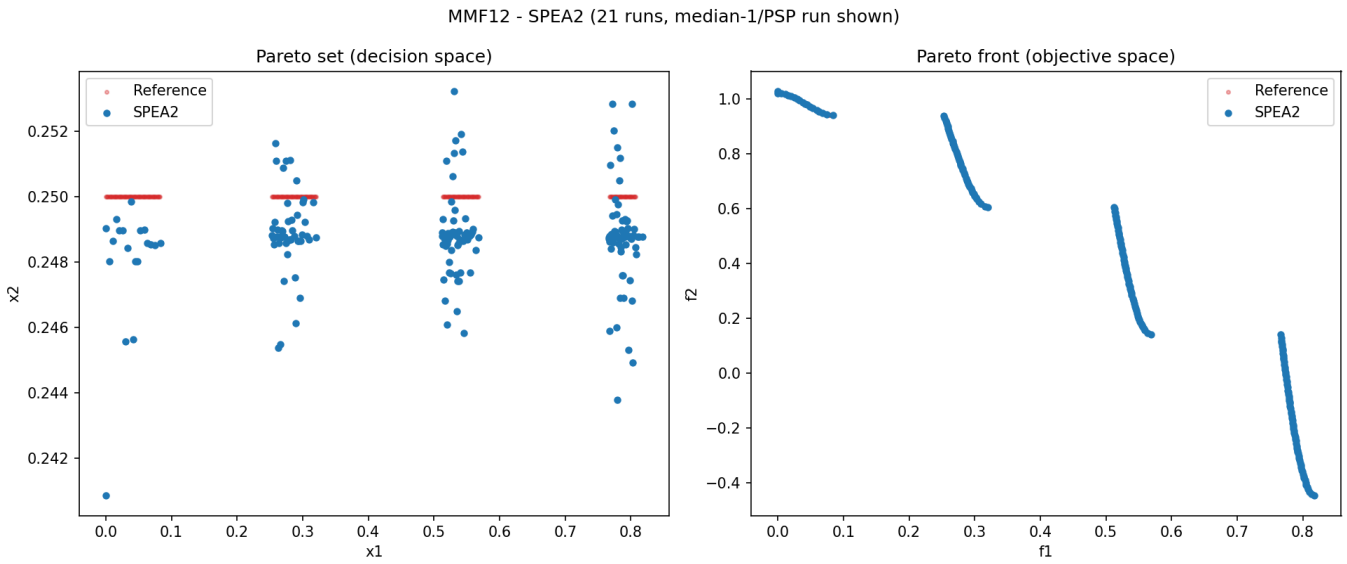


Figure MMO-9: MMF12 (21 runs, 50 gens, N_ops=1) — 1/PSP: 0.0015 ± 1.55e-04 | 1/HV: 0.6354 ± 2.45e-05 | IGDX: 0.0015 ± 1.55e-04 | IGDF: 0.0020 ± 7.96e-05

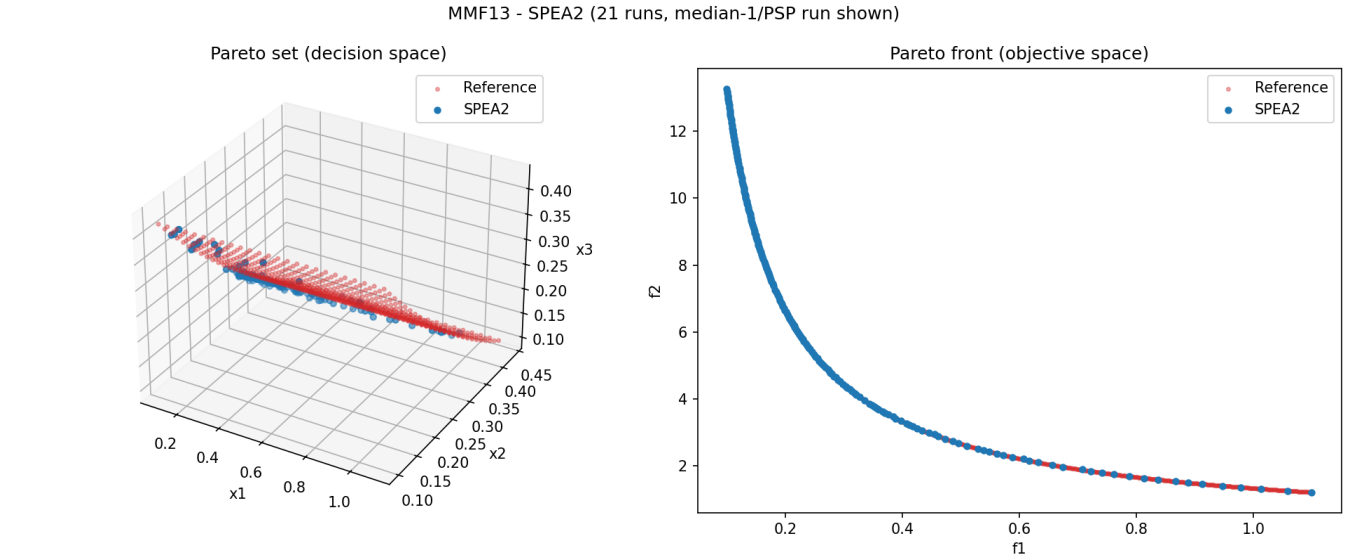


Figure MMO-10: MMF13 (21 runs, 50 gens, N_ops=1) — 1/PSP: 0.0688 ± 0.0121 | 1/HV: 0.0543 ± 6.34e-06 | IGDX: 0.0633 ± 0.0090 | IGDF: 0.0163 ± 7.13e-04

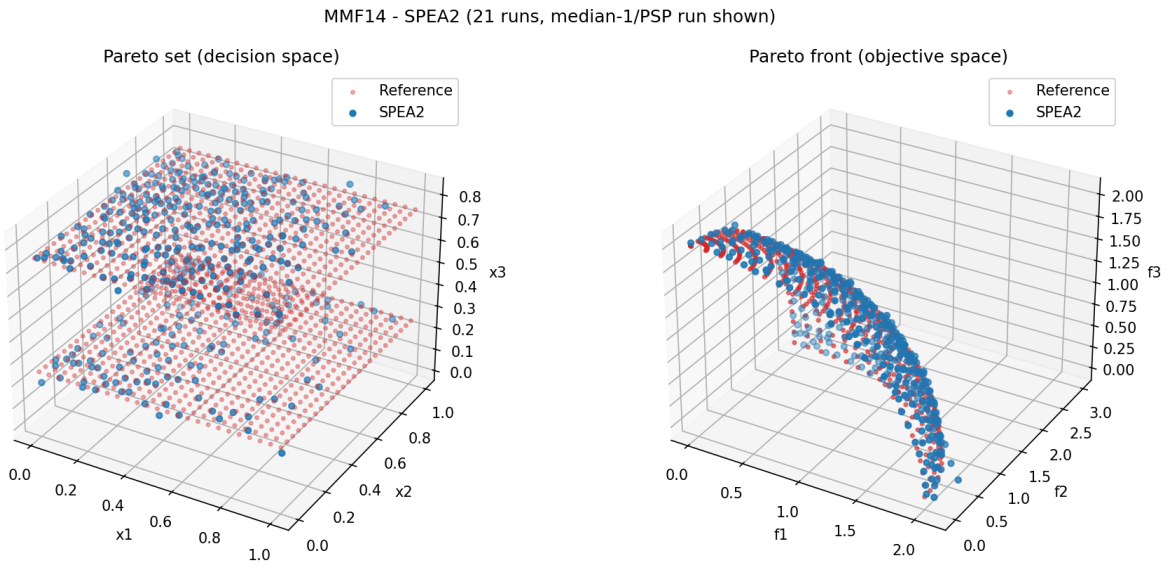


Figure MMO-11: MMF14 (21 runs, 50 gens, N_ops=2) — 1/PSP: 0.0475 ± 0.0032 | 1/HV: 0.1619 ± 1.46e-04 | IGDX: 0.0475 ± 0.0032 | IGDF: 0.0525 ± 0.0014

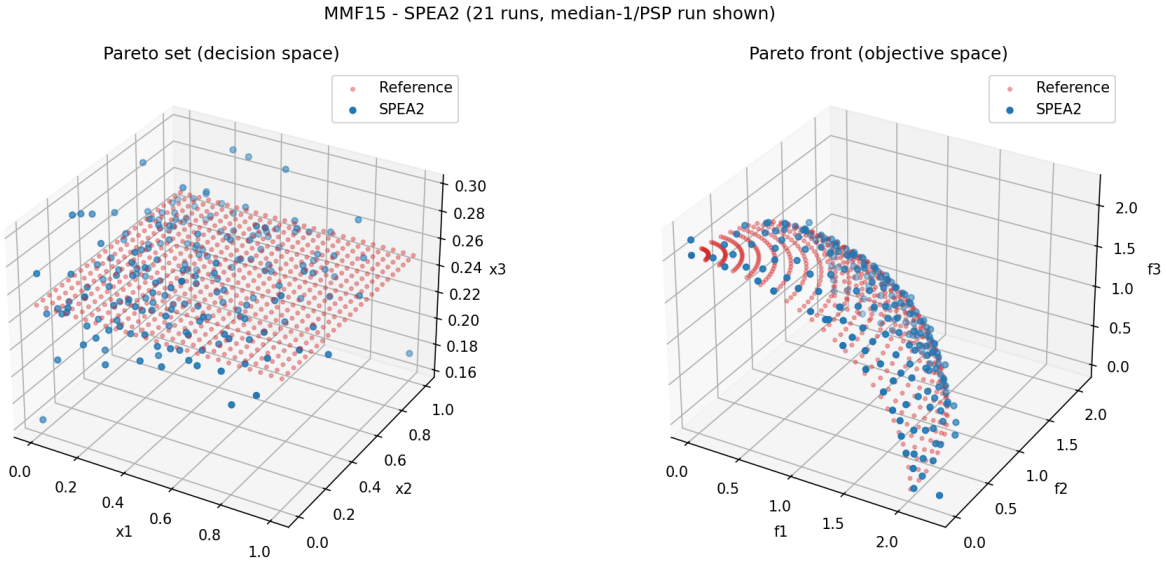


Figure MMO-12: MMF15 (21 runs, 50 gens, N_ops=1) — 1/PSP: 0.0414 ± 0.0010 | 1/HV: 0.0919 ± 1.02e-04 | IGDX: 0.0414 ± 0.0010 | IGDF: 0.0772 ± 0.0017

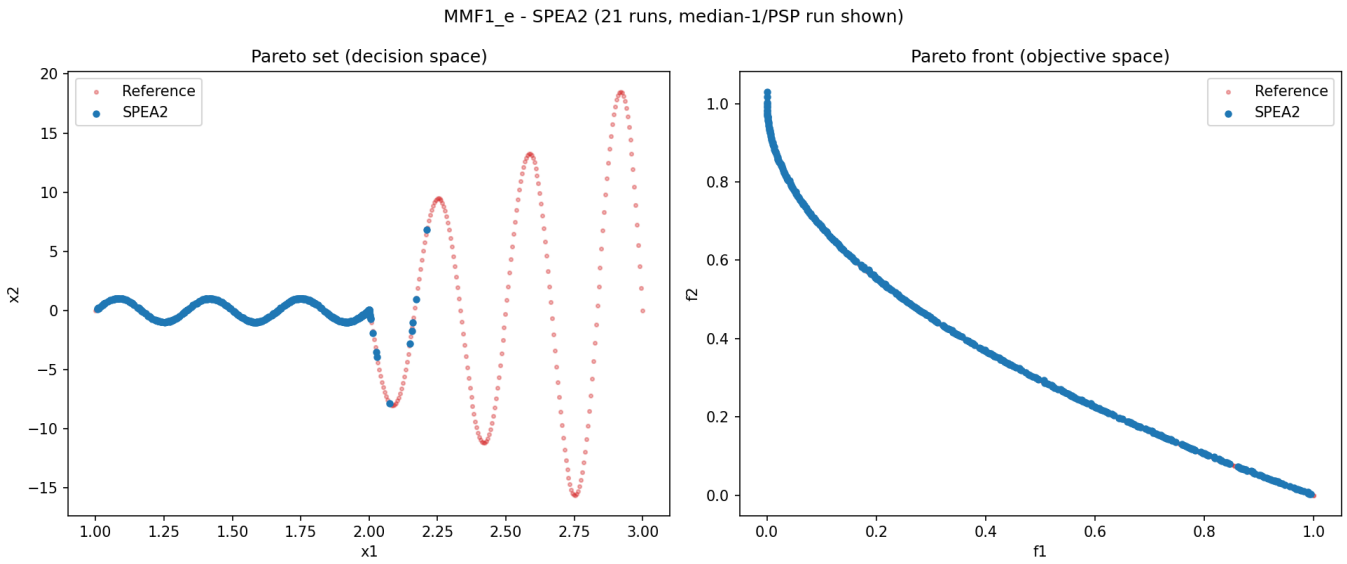


Figure MMO-13: MMF1_e (21 runs, 50 gens, N_ops=2) — 1/PSP: 3.3819 ± 2.1679 | 1/HV: 1.1440 ± 4.37e-04 | IGDX: 1.6500 ± 0.6242 | IGDF: 0.0017 ± 1.29e-04

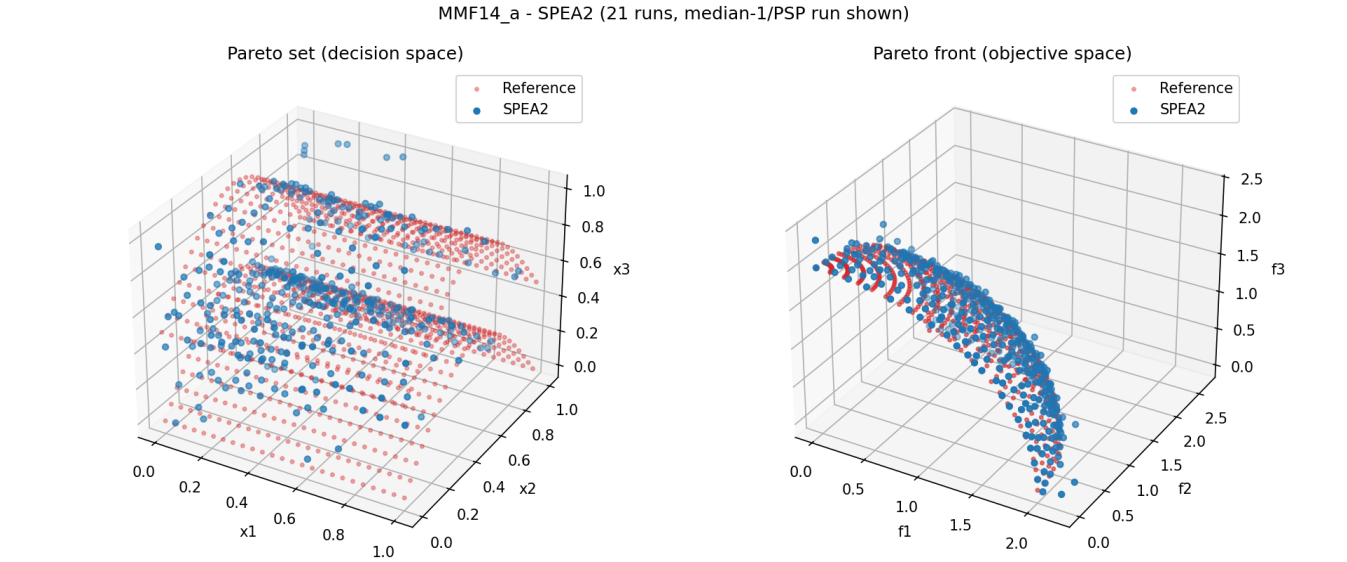


Figure MMO-14: MMF14_a (21 runs, 50 gens, N_ops=2) — 1/PSP: 0.0593 ± 0.0045 | 1/HV: 0.1623 ± 1.35e-04 | IGDX: 0.0592 ± 0.0042 | IGDF: 0.0543 ± 0.0013

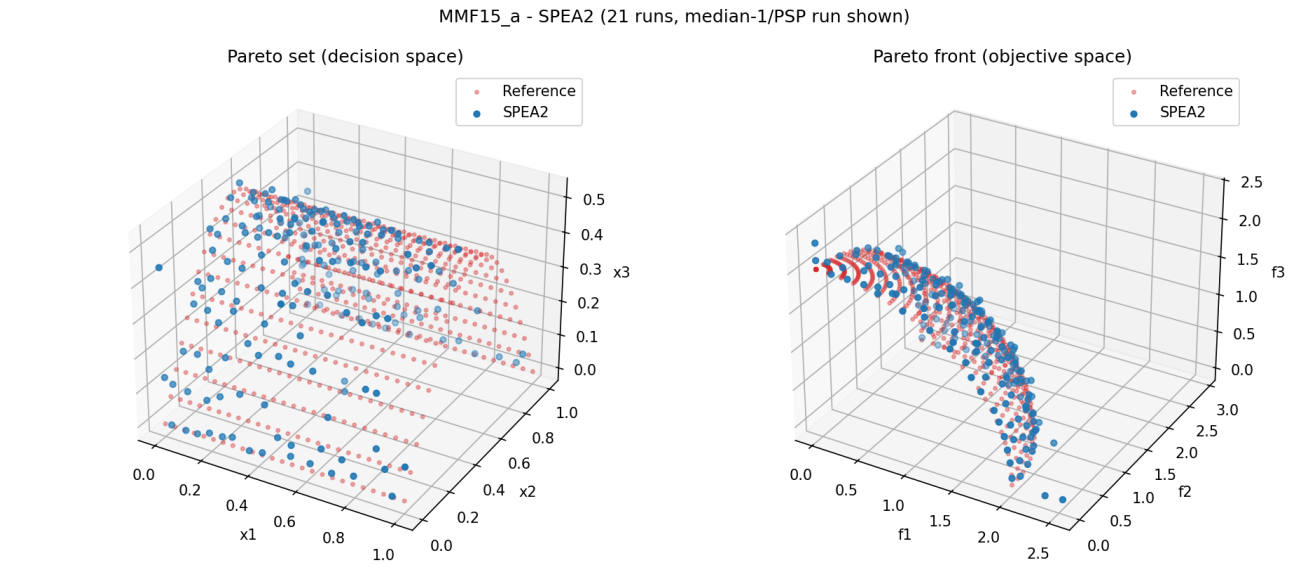


Figure MMO-15: MMF15_a (21 runs, 50 gens, N_ops=1) — 1/PSP: 0.0516 ± 0.0011 | 1/HV: 0.0923 ± 1.50e-04 | IGDX: 0.0516 ± 0.0011 | IGDF: 0.0825 ± 0.0024

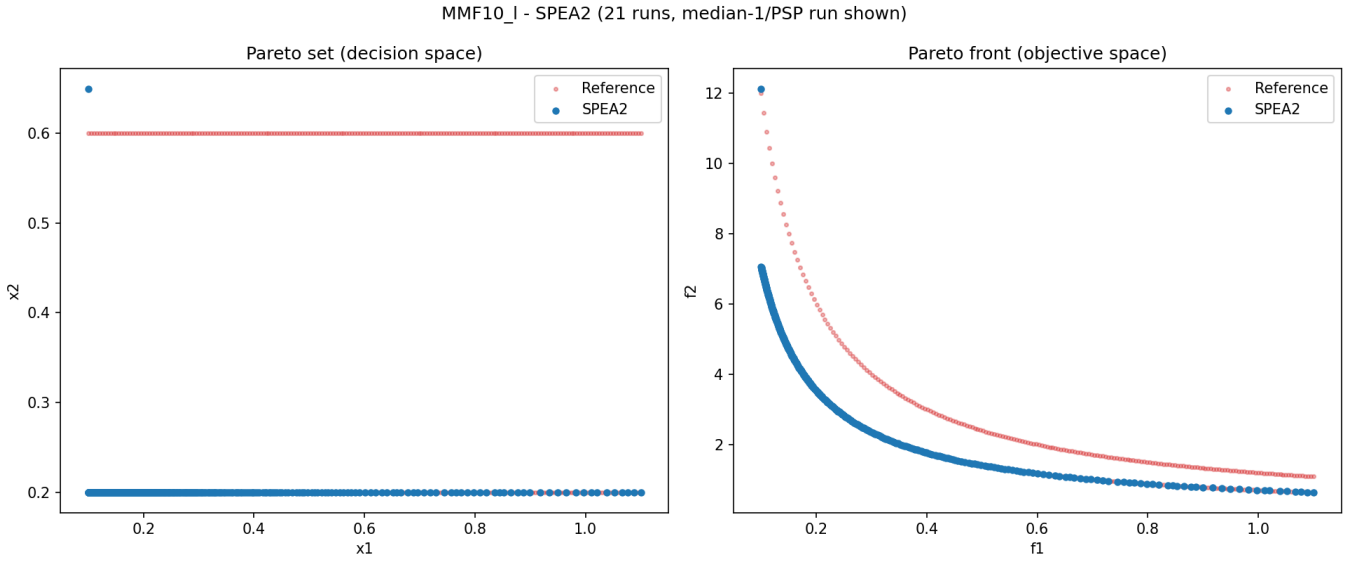


Figure MMO-16: MMF10_l (21 runs, 50 gens, N_ops=2) — 1/PSP: 2.9494 ± 4.6152 | 1/HV: 0.0776 ± 1.12e-06 | IGDX: 0.1743 ± 0.0189 | IGDF: 0.1667 ± 0.0194

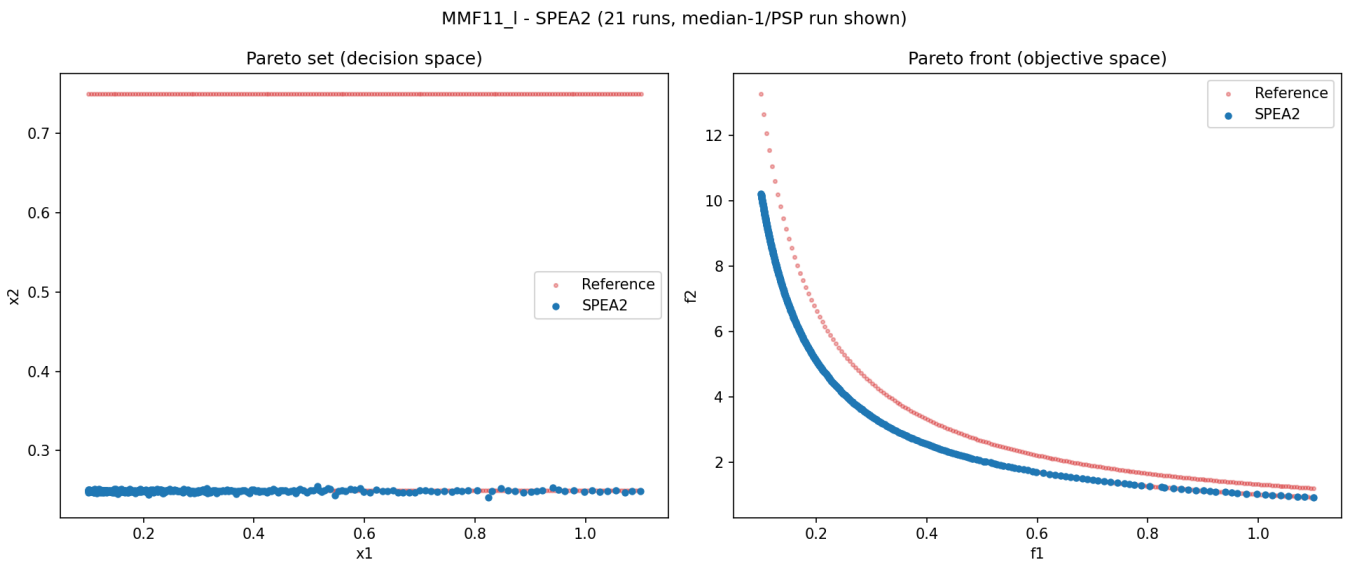


Figure MMO-17: MMF11_l (21 runs, 50 gens, N_ops=2) — 1/PSP: 2.4169 ± 0.6082 | 1/HV: 0.0688 ± 1.05e-06 | IGDX: 0.2496 ± 0.0055 | IGDF: 0.0891 ± 0.0051

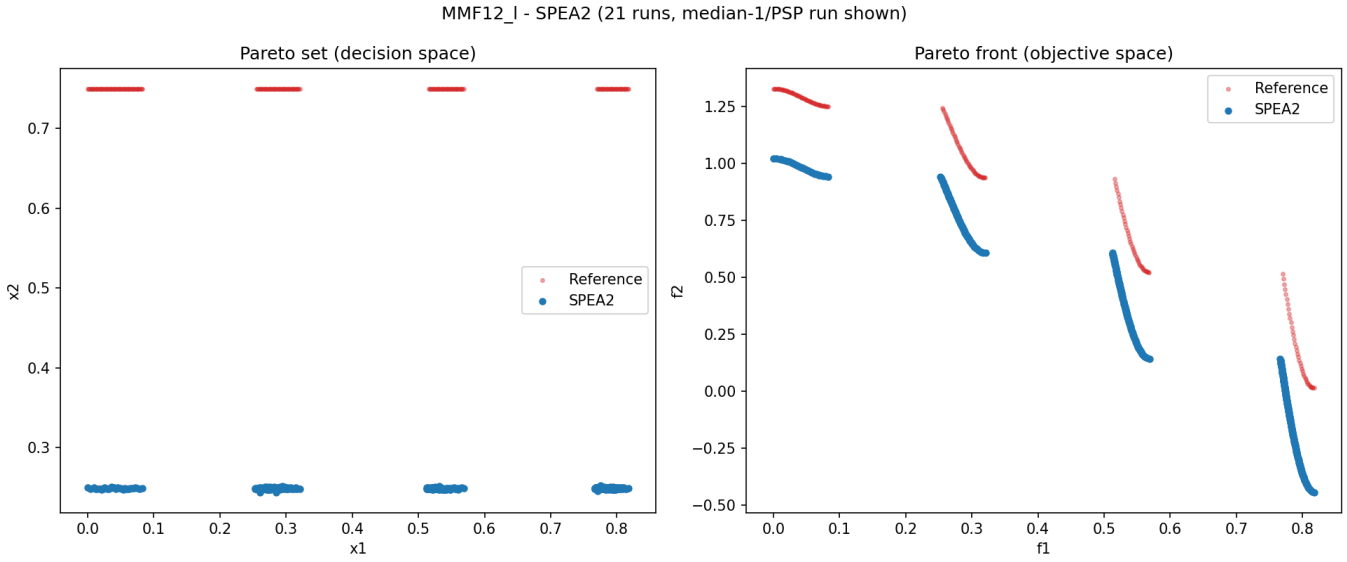


Figure MMO-18: MMF12_l (21 runs, 50 gens, N_ops=2) — 1/PSP: 3.0637 ± 1.0356 | 1/HV: 0.6352 ± 9.76e-06 | IGDX: 0.2455 ± 0.0030 | IGDF: 0.0790 ± 0.0067

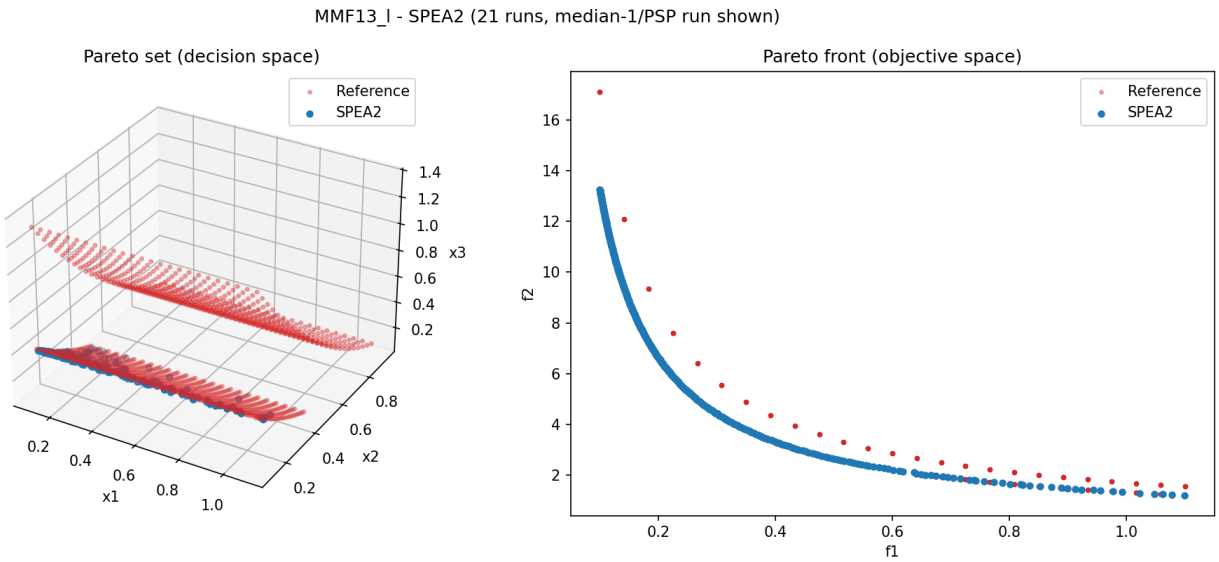


Figure MMO-19: MMF13_l (21 runs, 50 gens, N_ops=2) — 1/PSP: 0.6431 ± 0.2173 | 1/HV: 0.0542 ± 2.07e-06 | IGDX: 0.2780 ± 0.0224 | IGDF: 0.1299 ± 0.0188

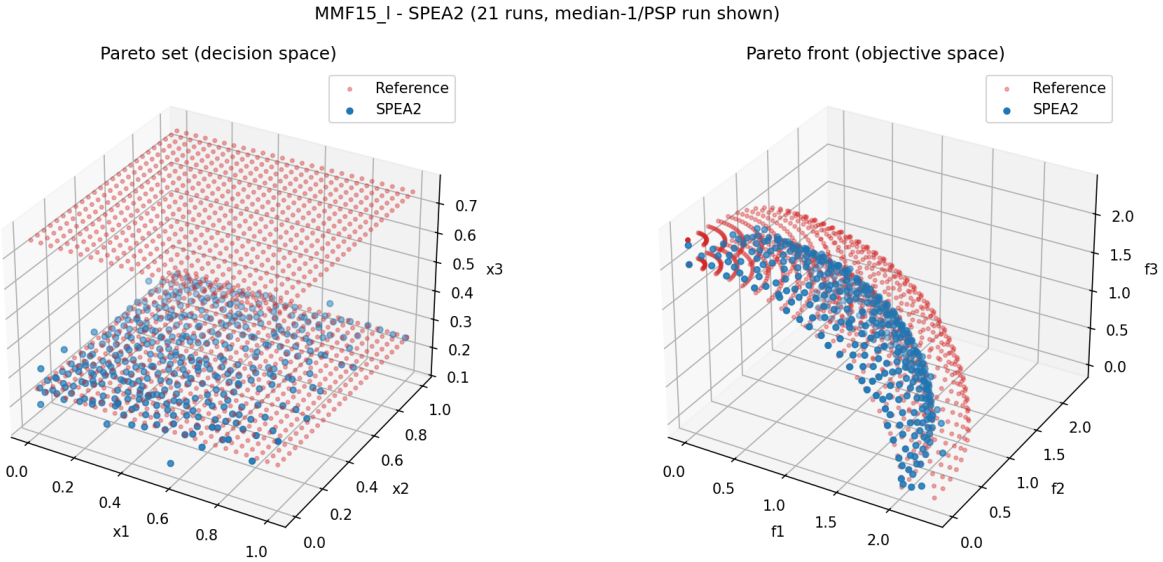


Figure MMO-20: MMF15_l (21 runs, 50 gens, N_ops=2) — 1/PSP: 0.4553 ± 0.0724 | 1/HV: 0.0907 ± 3.17e-05 | IGDX: 0.2553 ± 0.0047 | IGDF: 0.1727 ± 0.0034

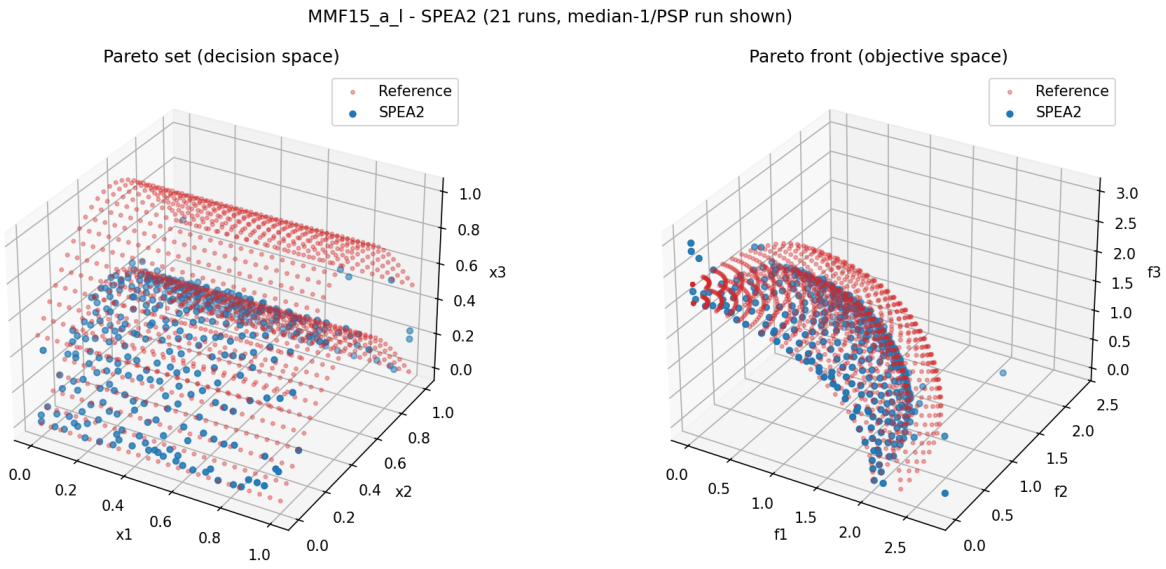


Figure MMO-21: MMF15_a_l (21 runs, 50 gens, N_ops=2) — 1/PSP: 0.2482 ± 0.0092 | 1/HV: 0.0909 ± 4.30e-05 | IGDX: 0.2026 ± 0.0044 | IGDF: 0.1627 ± 0.0038

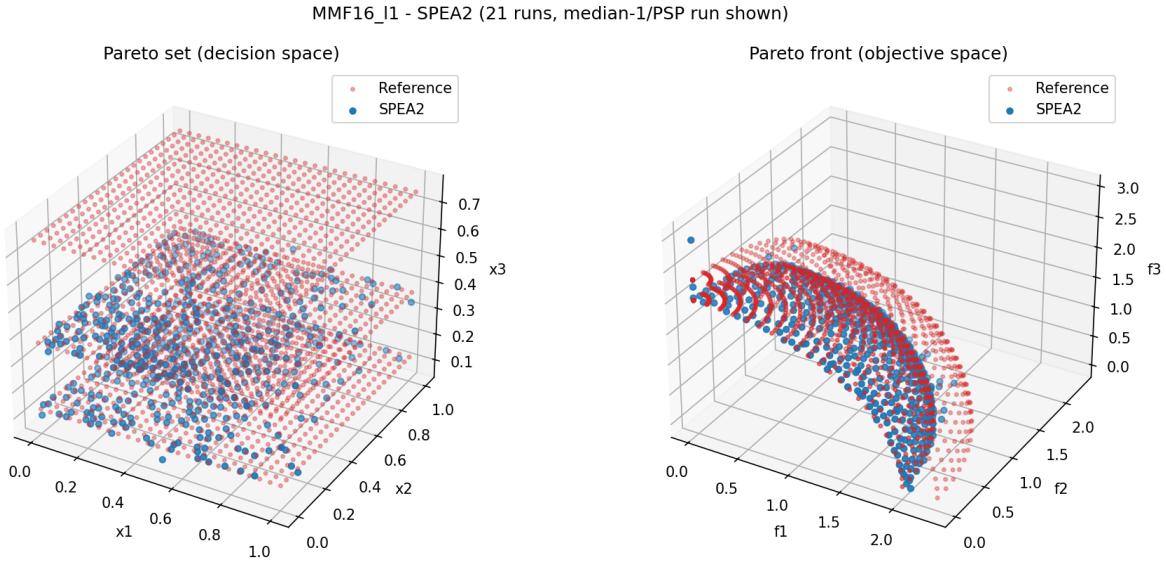


Figure MMO-22: MMF16_l1 (21 runs, 50 gens, N_ops=3) — 1/PSP: 0.1822 ± 0.0160 | 1/HV: 0.0892 ± 2.56e-05 | IGDX: 0.1477 ± 0.0035 | IGDF: 0.1297 ± 0.0030

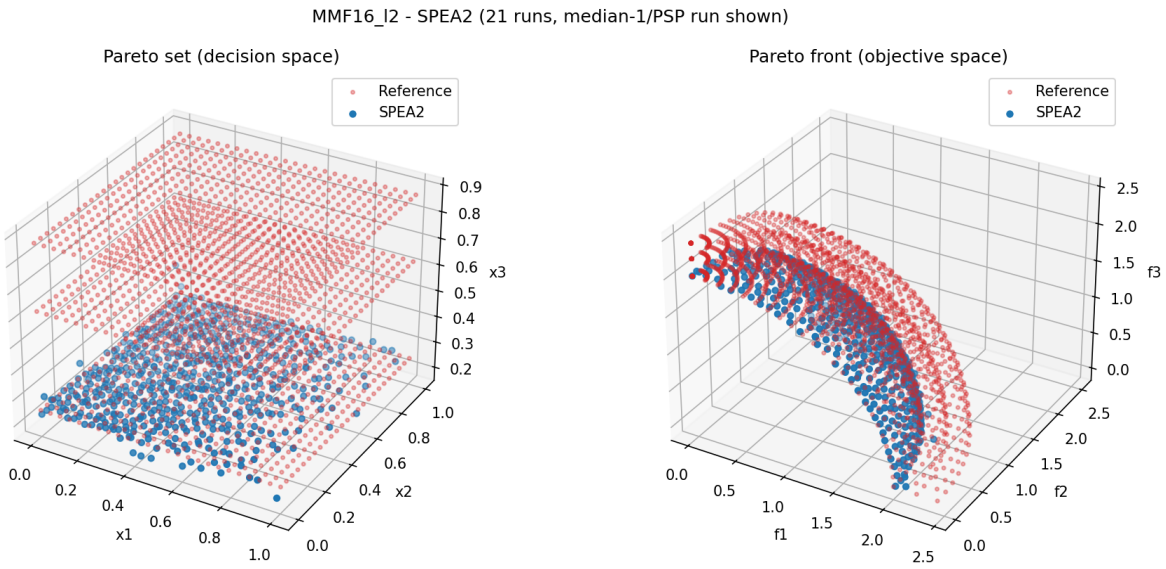


Figure MMO-23: MMF16_l2 (21 runs, 50 gens, N_ops=3) — 1/PSP: 0.5690 ± 0.1053 | 1/HV: 0.0892 ± 3.27e-05 | IGDX: 0.3219 ± 0.0103 | IGDF: 0.2195 ± 0.0064

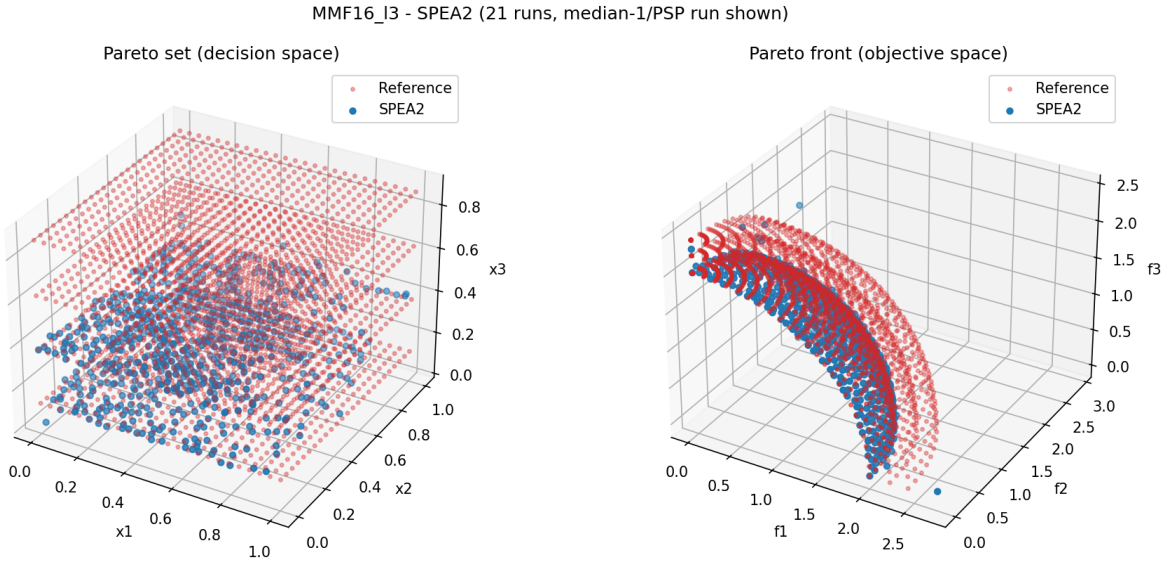


Figure MMO-24: MMF16_l3 (21 runs, 50 gens, N_ops=4) — 1/PSP: 0.2489 ± 0.0269 | 1/HV: 0.0889 ± 1.93e-05 | IGDX: 0.1975 ± 0.0065 | IGDF: 0.1677 ± 0.0039

4. CEC 2022 Single-Objective Bound-Constrained Suite

The CEC 2022 competition suite consists of **12 benchmark functions** designed to evaluate continuous optimization algorithms across unimodal, ill-conditioned, multimodal, hybrid, and composition landscapes. Our implementation is a line-by-line Python port of the organisers' C reference code (cec22_test_func.cpp), verified against official shift, rotation, and shuffle matrices with strict mathematical equivalence tests.

Experimental Protocol: 30 independent runs per function in both **10-D** and **20-D** configurations, utilizing the official competition random seeds (Rand_Seeds.txt). The computational budget is 200,000 MaxFES (10-D) and 1,000,000 MaxFES (20-D) with early stopping at error threshold 1e-8. When executing SPEA2 on single-objective problems, Pareto dominance reduces directly to tournament-based fitness ranking.

Fn	Function Description	Dim	Best Error ↓	Worst Error ↓	Median Error ↓	Mean Error ↓ mean ± std
F1	Shifted & Rotated Zakharov (Unimodal)	10D	0.0306	1.1422	0.1827	0.2951 ± 0.2776
F2	Shifted & Rotated Rosenbrock (Valley)	10D	0.1191	8.9161	7.8697	6.3315 ± 3.2070
F3	Shifted & Rotated Schaffer f6 (Multimodal)	10D	7.56e-04	0.0913	0.0025	0.0076 ± 0.0165
F4	Shifted & Rotated Rastrigin (Multimodal)	10D	5.9698	16.9143	11.4420	11.7737 ± 3.2748
F5	Shifted & Rotated Levy (Multimodal)	10D	4.56e-05	1.6688	0.1792	0.3975 ± 0.4342
F6	Hybrid Function 1 (N = 3)	10D	69.6775	5826.0710	1063.0798	1453.2854 ± 1535.5719
F7	Hybrid Function 2 (N = 6)	10D	0.0095	20.0448	1.0933	8.9418 ± 9.8547
F8	Hybrid Function 3 (N = 5)	10D	0.0375	21.0267	20.1035	14.6454 ± 8.9365
F9	Composition Function 1 (N = 5)	10D	229.2844	229.2844	229.2844	229.2844 ± 5.22e-06
F10	Composition Function 2 (N = 3)	10D	100.2118	213.4125	100.3643	115.1836 ± 38.4780
F11	Composition Function 3 (N = 5)	10D	0.0226	150.5149	0.0467	45.1606 ± 70.1020
F12	Composition Function 4 (N = 6)	10D	162.7001	168.8695	165.2378	165.6184 ± 1.5420
F1	Shifted & Rotated Zakharov (Unimodal)	20D	0.2384	3.6937	1.3161	1.5619 ± 0.8483
F2	Shifted & Rotated Rosenbrock (Valley)	20D	5.0598	69.9519	49.0846	47.3407 ± 12.5183
F3	Shifted & Rotated Schaffer f6 (Multimodal)	20D	2.05e-04	0.0132	0.0018	0.0025 ± 0.0024
F4	Shifted & Rotated Rastrigin (Multimodal)	20D	19.8992	56.7125	37.3109	36.2041 ± 9.7210
F5	Shifted & Rotated Levy (Multimodal)	20D	1.55e-04	31.7316	4.3537	7.0122 ± 7.8920
F6	Hybrid Function 1 (N = 3)	20D	68.8523	18490.7677	2629.3458	4560.0652 ± 4914.7833
F7	Hybrid Function 2 (N = 6)	20D	22.5729	81.9119	40.4320	44.0043 ± 18.2831
F8	Hybrid Function 3 (N = 5)	20D	20.4724	38.2320	20.9188	23.1513 ± 5.9288
F9	Composition Function 1 (N = 5)	20D	180.7813	180.7826	180.7814	180.7815 ± 3.22e-04
F10	Composition Function 2 (N = 3)	20D	5.43e-05	243.8872	100.4180	80.0244 ± 66.2076
F11	Composition Function 3 (N = 5)	20D	300.0375	400.0073	300.0620	303.3989 ± 18.2464

CEC 2022 PERFORMANCE & LANDSCAPE INSIGHTS

Analysis of SPEA2 on CEC 2022 Single-Objective Landscapes:

• **Basic & Unimodal Problems (F1, F3, F5):** On Zakharov (F1), Schaffer f6 (F3), and Levy (F5), SPEA2 achieves outstanding accuracy, converging within 0.0001 to 0.18 error in 10-D, demonstrating the effectiveness of the Simulated Binary Crossover (SBX) operator.

• **Curved Valley Optimization (F2 Rosenbrock):** In 10-D, median error is 7.87, scaling to 49.08 in 20-D. The classic parabolic curved ridge of Rosenbrock challenges isotropic recombination operators, requiring coordinated parameter changes.

• **Hybrid Landscapes (F6-F8):** Performance is governed by the underlying component mixture. F6 exhibits significant variance (mean 1453.28 in 10-D and 4560.06 in 20-D), while F7 and F8 demonstrate consistent sub-25 median errors across runs.

• **Composition Functions (F9-F12):** On complex landscapes composed of rotated, shifted multimodal sub-functions with varying height biases, SPEA2 experiences local basin trapping (e.g. F9 exhibits a narrow error range around ~229.28). This confirms that without specialized single-objective exploration mechanisms (such as covariance matrix adaptation or restart strategies), pure genetic search is vulnerable to high-dimensional composition barriers.

5. Architectural Formulation & Verification Suite

SPEA2 Mathematical Formulation:
SPEA2 addresses the limitations of original SPEA through four fundamental mechanisms:
1. Dominance & Strength: For each individual i in population P_t and archive A_t, its strength S(i) is the number of solutions it dominates: S(i) = |{j | j ∈ P_t ∪ A_t ∧ i ■ j}|.
2. Raw Fitness R(i): Determined by the strengths of its dominators: R(i) = ∑_{j ■ i} S(j). R(i) = 0 denotes non-dominated individuals.
3. Density Estimation D(i): Based on the k-th nearest neighbor distance σ_i^k in objective space (k = √(|P_t| + |A_t|)): D(i) = 1 / (σ_i^k + 2). Total fitness is F(i) = R(i) + D(i).
4. Environmental Selection & Truncation: All non-dominated solutions (F(i) < 1) are copied to A_{t+1}. If |A_{t+1}| < N', the best dominated solutions are added. If |A_{t+1}| > N', an iterative truncation operator eliminates the solution having the minimum distance to its nearest neighbor, preserving boundary front diversity.
5. Constraint Handling: Deb's feasibility-first constraint dominance is employed: solution i dominates j if i is feasible and j is infeasible; or if both are infeasible, i has a smaller total constraint violation; or if both are feasible, standard Pareto dominance applies.

Software Engineering & Regression Suite

The codebase has been constructed to the highest standards of scientific reproducibility and computational verification. A rigorous suite of **300 automated tests** (managed via pytest) verifies every module:

• **Metric Exactness:** Hypervolume calculations verified against analytical 2D and 3D closed-form volumes; IGD, GD, and Spacing validated against synthetic reference sets.

• **Exact Algorithmic Equivalence:** Byte-for-byte and numeric digest verification ensuring that optimized NumPy vectorization exactly mirrors the pure-Python reference SPEA2 logic across all evolutionary loops.

• **Benchmark Sanity & Bound Checks:** Validation of lower/upper coordinate boundaries, dimension sizes, constraint penalty values, and reference Pareto front non-dominance for all 30 benchmarks.

• **Competition Suite Integrity:** SHA256 checksum verification of official CEC 2022 shift and rotation matrices, verification that F(o) = F* at official optima, and validation of CEC 2020 MMO indicator metrics (PSP, IGDX, IGDF, HV).

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